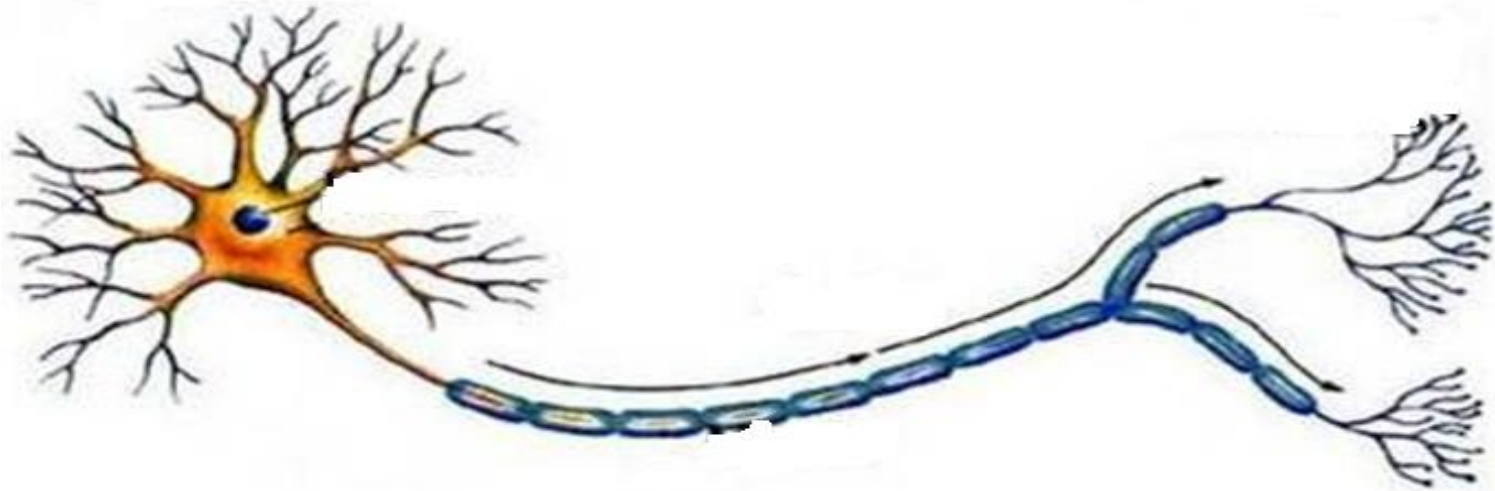


Physiology of Excitable Tissues



Neurophysiology

Outline

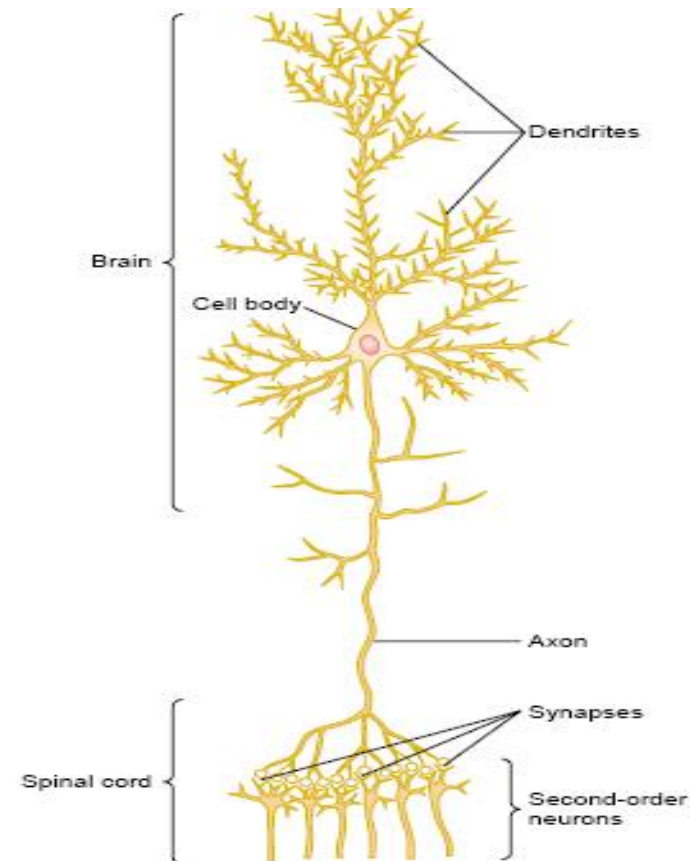
- Functional structure of neurons
- Classification of neurons and neuroglia cells
- Membrane potential (resting membrane potential)
 - Graded potential
 - Action potential (nerve impulse)
 - Propagation of nerve impulse
- Synapses
 - Synaptic transmission

Nerve Cells

- The nervous system have two principal cell types
 1. Neurons
 2. Neuroglial cells

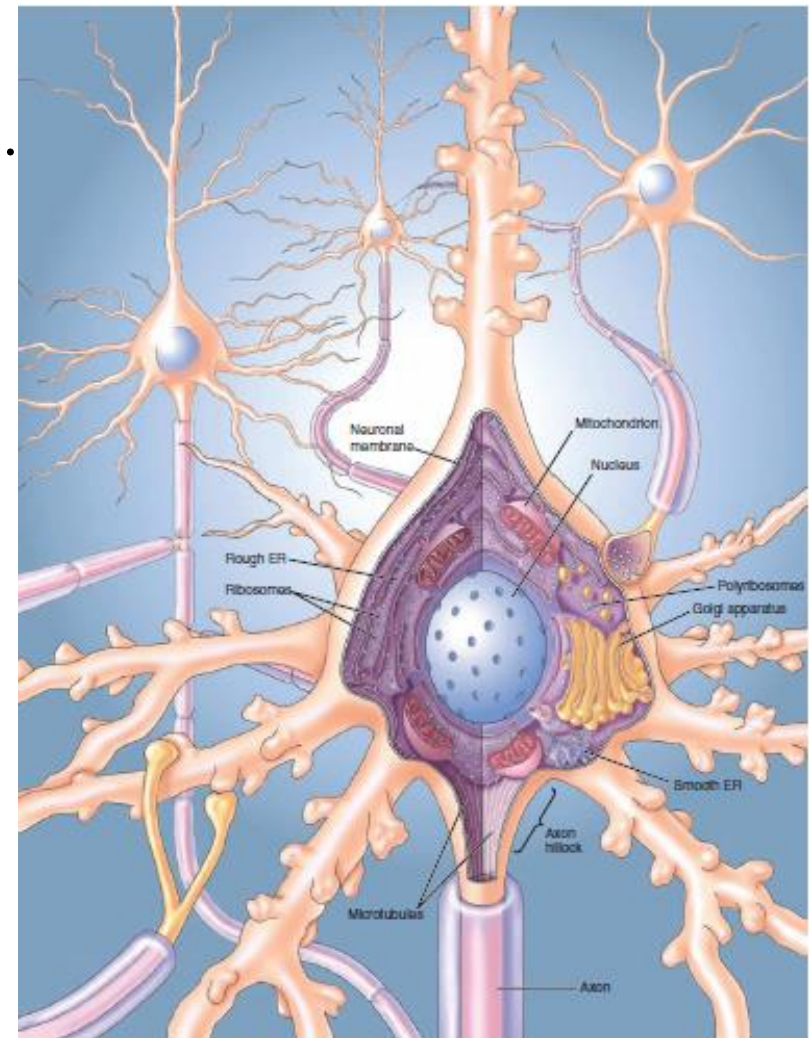
Neurons

- Structural and functional unit of the nervous system
- Human brain: $\approx 10^{11}$ neurons & 10^{14} synapses
- Structural components:
 - Soma (cell body)
 - Dendrites
 - Axon
 - Presynaptic Terminal



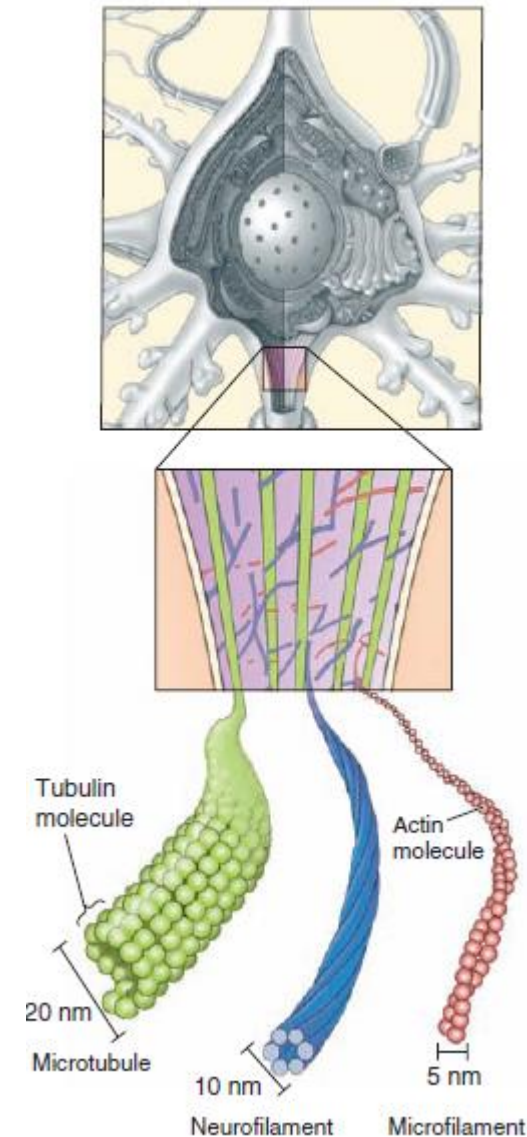
Soma (Cell Body)

- Gives rise to axon and dendrites.
- Has nucleus, nucleolus, mitochondria ...
- Nissl bodies: RER + prominent GA.
- Cytoskeletal elements:
 - Microtubules
 - Neurofilaments
 - Microfilaments



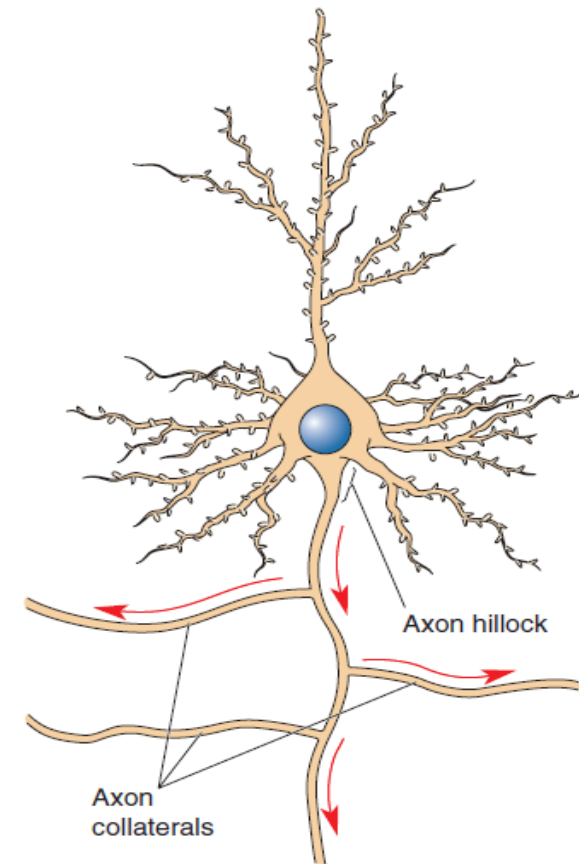
Soma (Cell Body)

- Functions:
 - Metabolic center of the neuron
 - Membrane constituents
 - Enzymes
 - Neurotransmitters (neuropeptides)
 - Reception and integration of incoming signals



Dendrites

- Origin: apical or basal, $\leq 1\text{mm}$ long
- Components:
 - Microtubules and neurofilaments
 - Proximal dendrites: Nissl's bodies + GA
 - Spines ($\uparrow$ SA, actin microfilaments (motile))

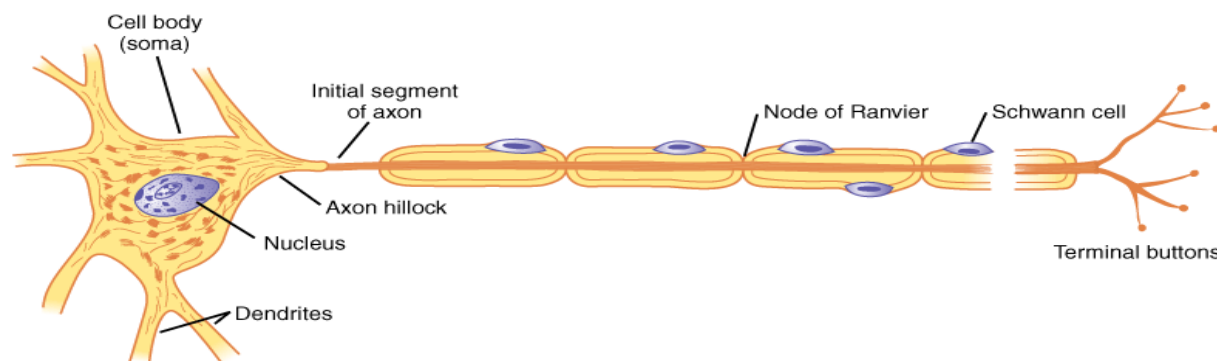


Dendrites

- *Functions:*
- Receive input signal from other neurons (*dendritic spines/tree*)
 - *90% surface area (synaptic contacts: $10^4 - 4 \times 10^5$)*
 - *Intelligence Vs. mental retardation (loss of dendritic spines)*
(ageing, Down syndrome)
- Computation or integration of the signal

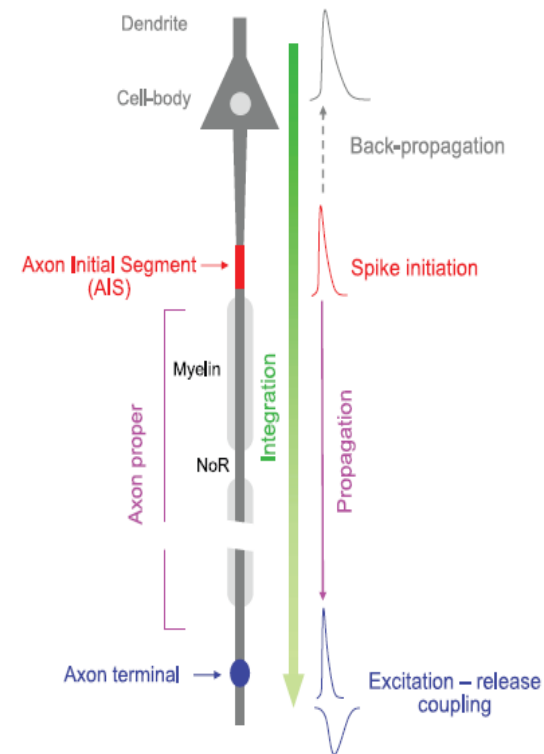
Axon

- Origin: soma, 1m long, **ONLY ONE/neuron**, uniform diameter
- Components: SER, prominent cytoskeleton(*Lacks RER/Nissl sub. free ribosomes and GA*)+ Mitochondria.
- Special features:
 - *Axon hillock*
 - Myelin sheath (+ *axolemma* + *axoplasm*)
 - *Nodes of Ranvier*



Axon

- Functions:
- Initiation of action potential at the axon hillock
 - High density of voltage-gated ion channels of Na^+ , K^+ , $\text{Ca}^{2+} \rightarrow \downarrow$ Threshold (-45mV)
- Impulse conduction: *action potential* (6-120m/s)

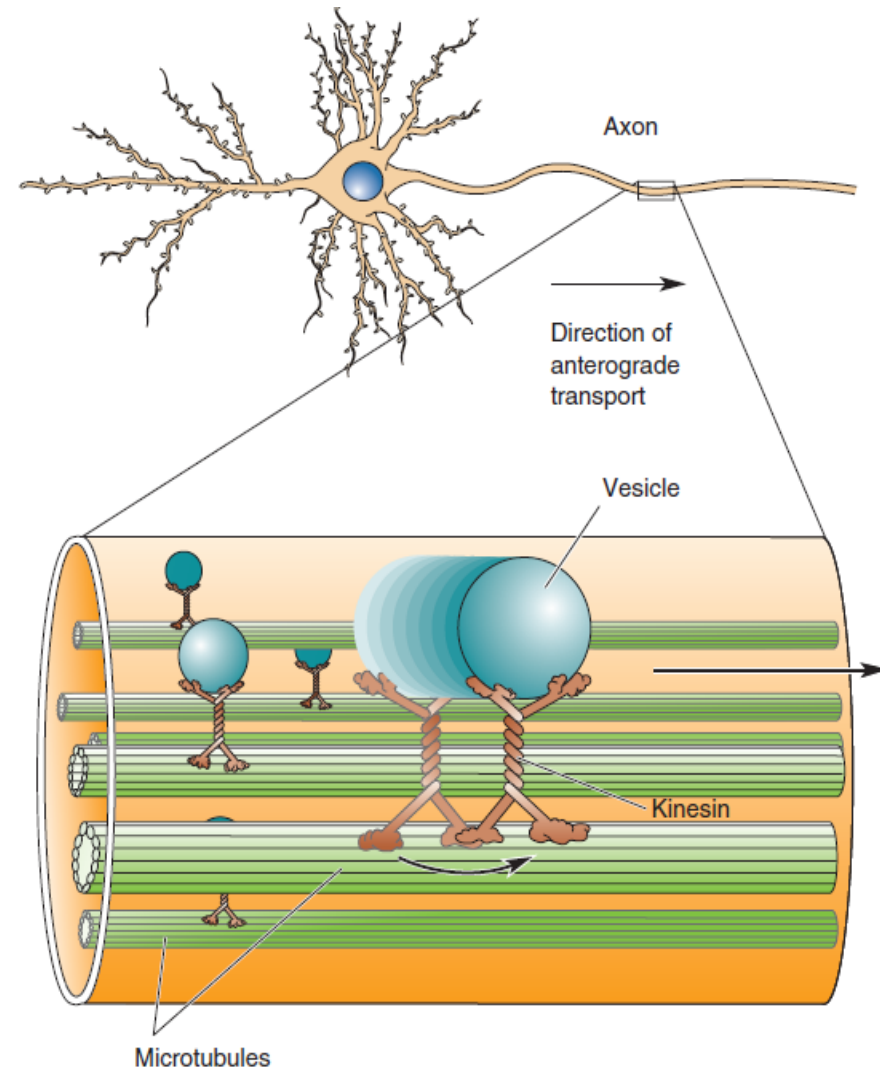


Axon

- Functions:
- Axoplasmic Transport
 - Occurs in the microtubules → ATP and Ca^{2+} dependent.
 - Purpose:
 - Replenishment of synaptic vesicles and enzymes
 - Responsible for NT transport
 - Bidirectional:
 - Anterograde: Soma/Cell body → Nerve terminal
 - Retrograde: Soma/Cell body ← Nerve terminal

Axon

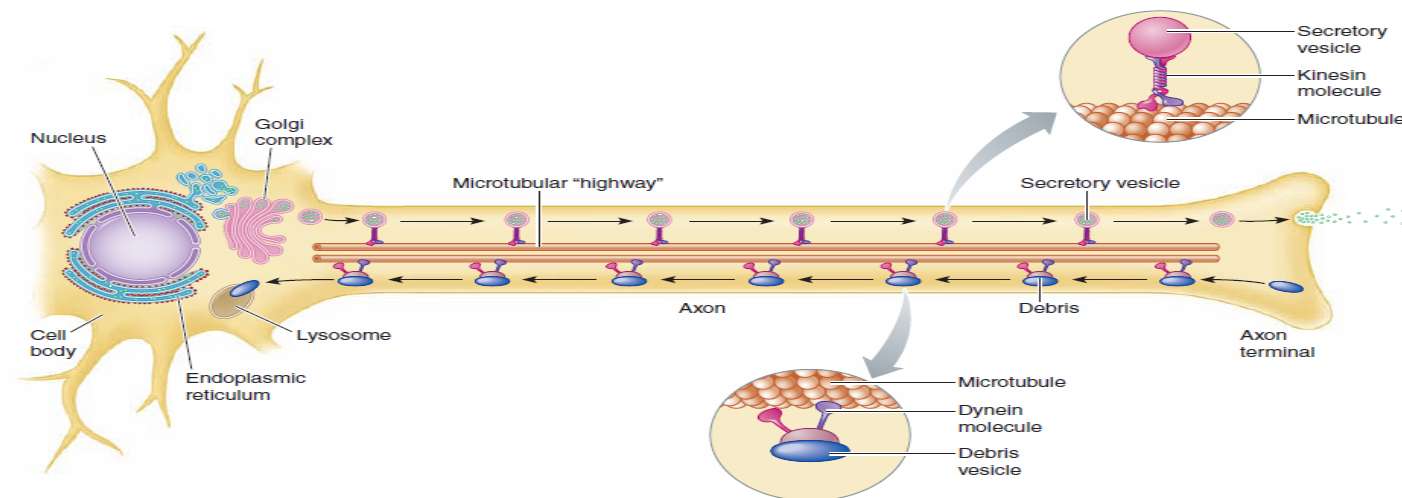
- Anterograde Axoplasmic Transport
- 2 types
 1. Fast axonal transport
 - 400mm/d (= 40cm/d)
 - Membrane-bound organelles + mitochondria (+ vesicles).
 - Motor molecule is **Kinesin**



Axon

2. Slow axonal transport

- 1mm/d (R= 0.2-2.5mm/d)
- Soluble proteins, cytoskeletal elements, precursors ...(actins, tubulins...)
- Not dependent on microtubule, no ATPase and motor molecule.

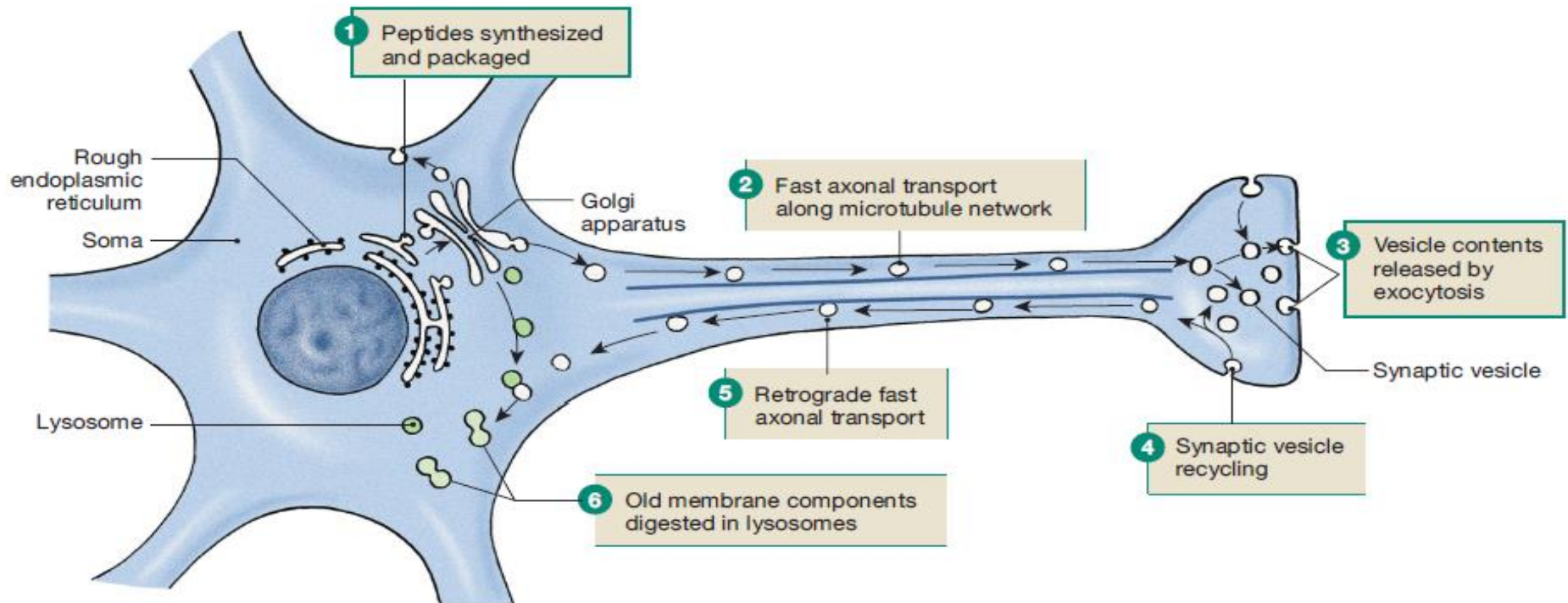


Axon

- Retrograde Axoplasmic Transport
 - Fast retrograde axonal transport
 - Velocity = 200 - 300mm/d (= 20cm/d - 30cm/d)
 - Motor Molecule is **Dynein**
 - Returns synaptic vesicle membrane to the soma for lysosomal degradation or recycling
 - Permits communication between the synaptic terminal and the cell body
 - Neurotrophic factors: NGF, BDNF, NT-3, NT-4 NT-5...

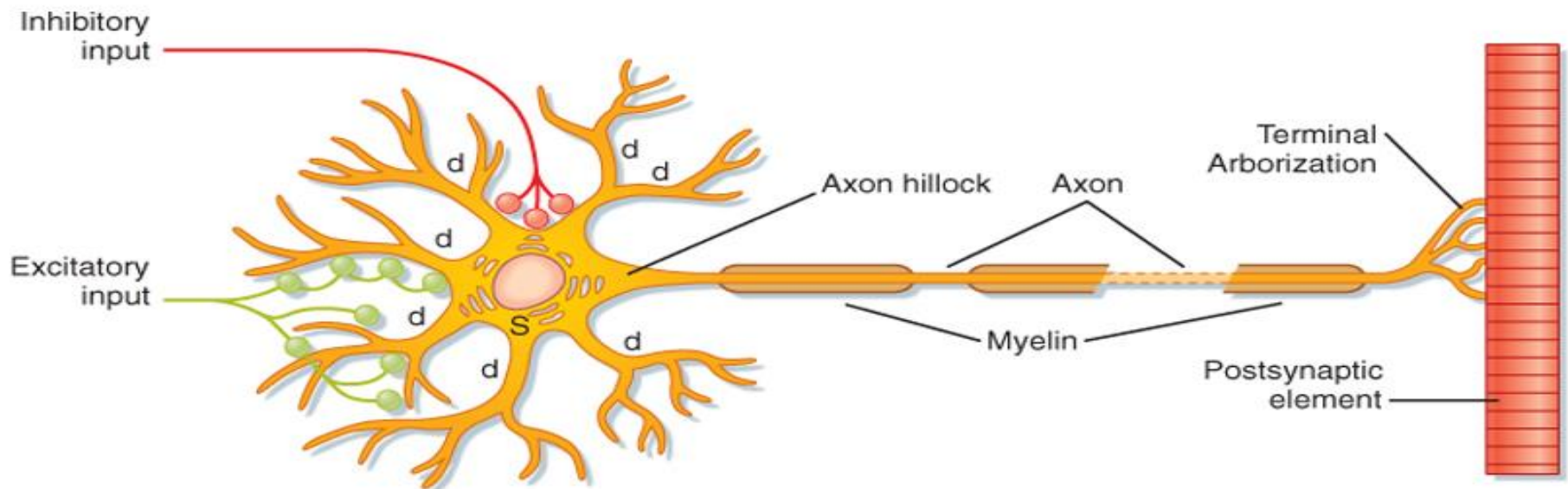
Axon

- Retrograde Axoplasmic Transport
 - Neurotropic viruses: *poliomyelitis virus*, *HZ*, *HSV*, *rabies virus*
 - Toxins: *tetanus toxin*

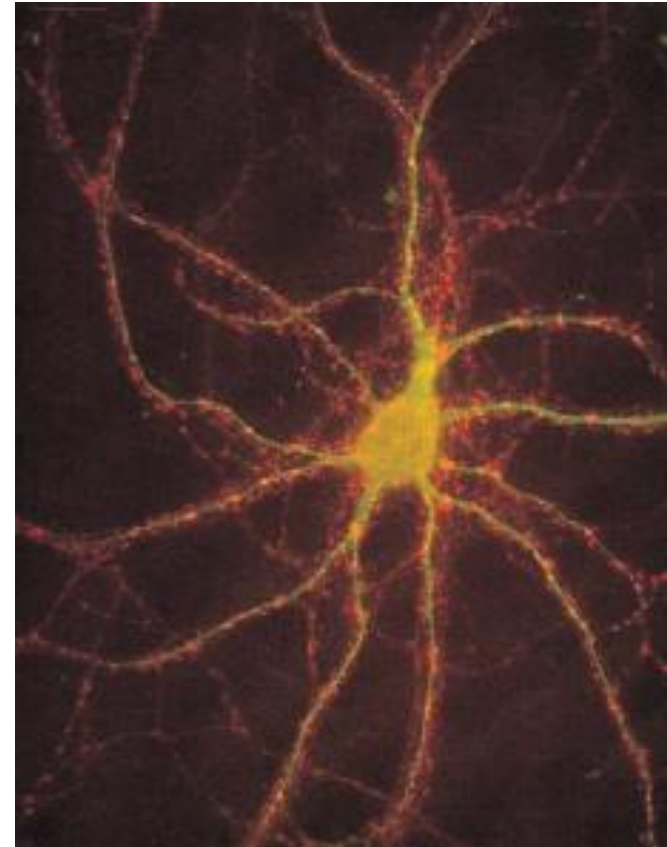
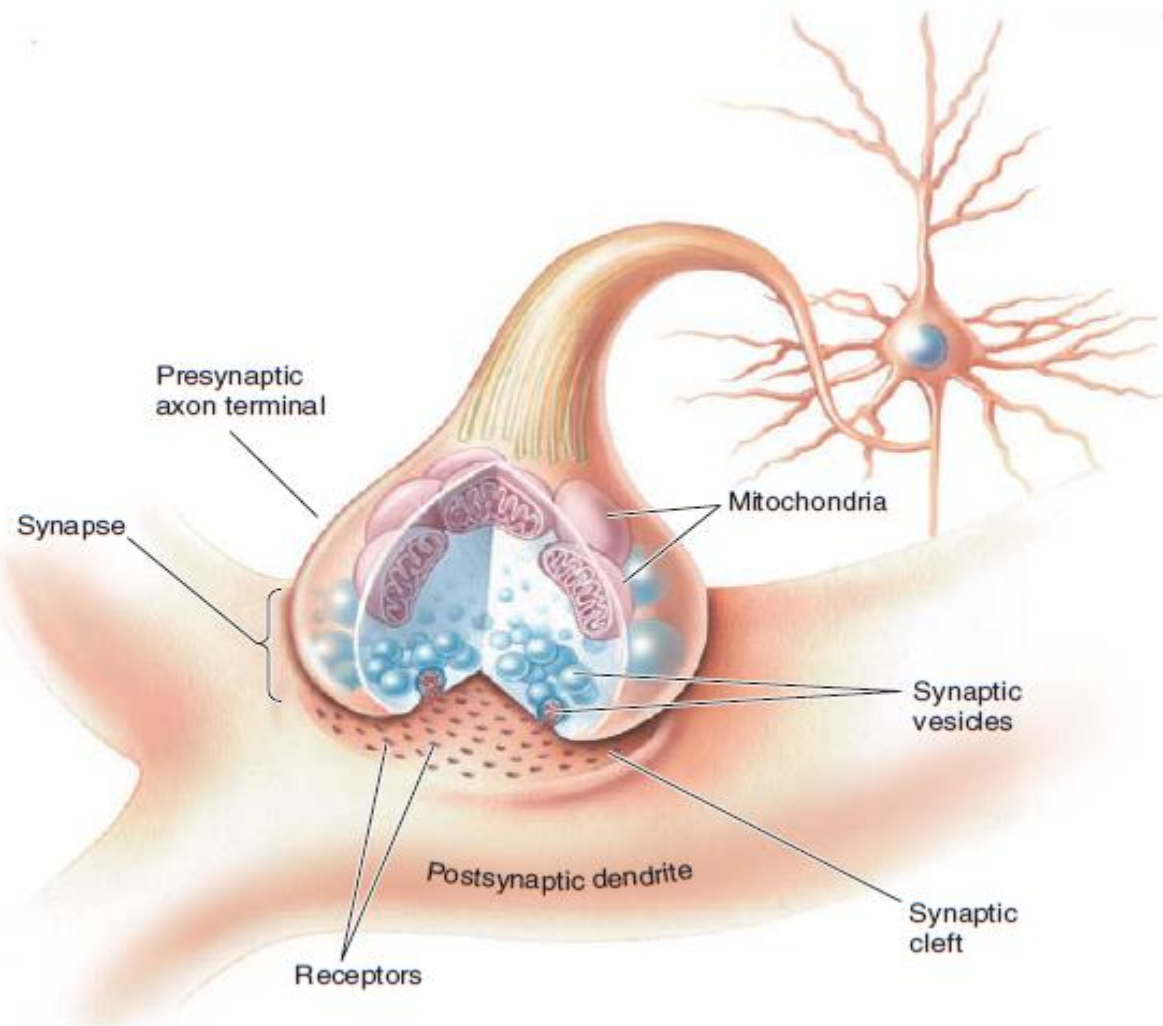


Synaptic Terminals

- Transmitting elements of the neuron.
- Synaptic vesicles, and high number of mitochondria
- The cell sending out information → Presynaptic cell
- The cell receiving the information → Postsynaptic cell



Synaptic Terminals



Classification of Neurons

A. On the basis of function

1. Afferent Neurons (*Sensory Neurons*)

- Transmit information into the CNS from receptors at their peripheral endings.
- Have no dendrites.
- Cell body + long peripheral processes are outside the CNS.
- Only the short central process enters the CNS.

Classification of Neurons

2. Efferent Neurons (*Motor Neurons*)

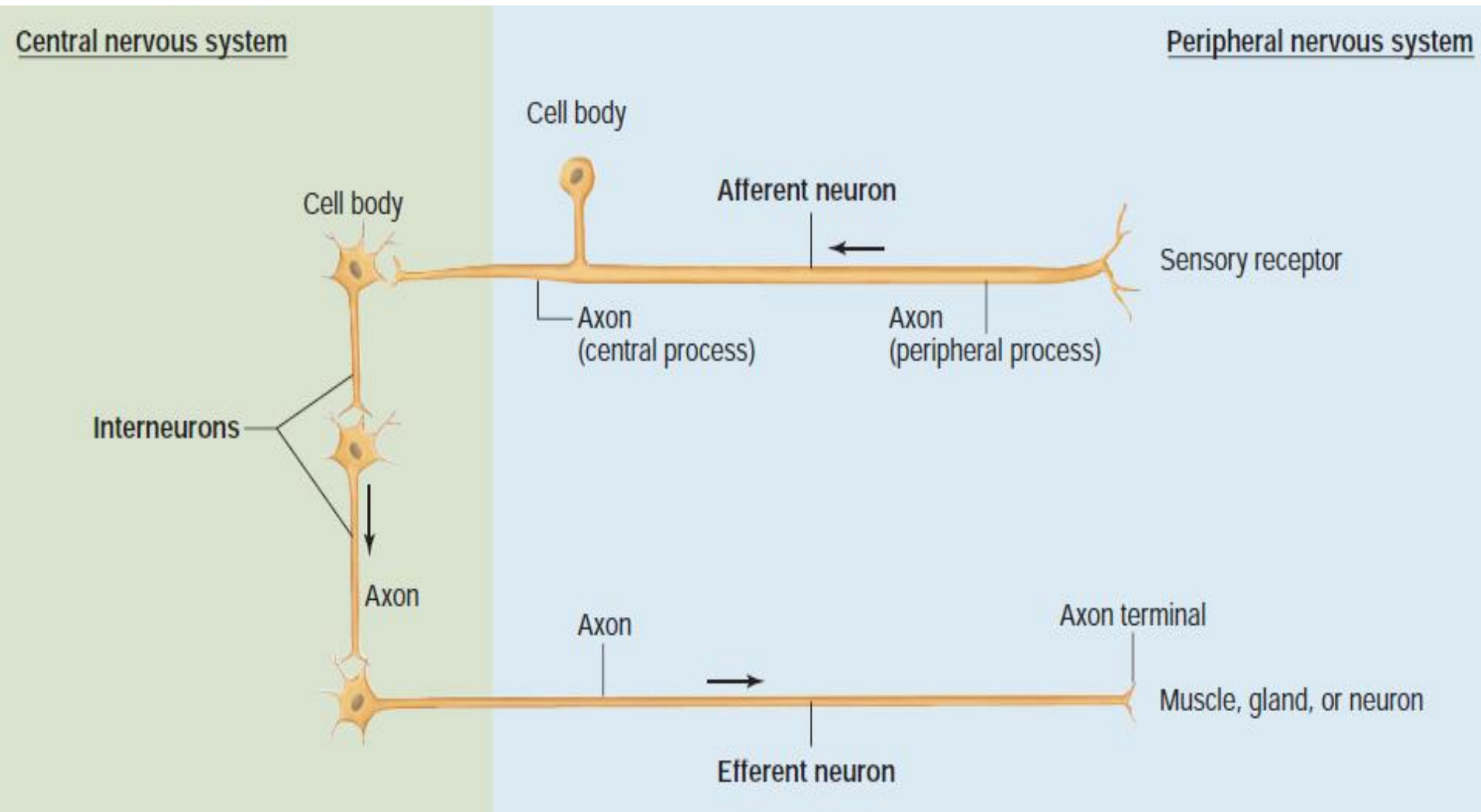
- Transmit information out of the CNS to effectors (neurons, muscles or glands)
 - Cell body + dendrites + small segment of the axon, in CNS.
 - Most of the axon is outside the CNS.

Classification of Neurons

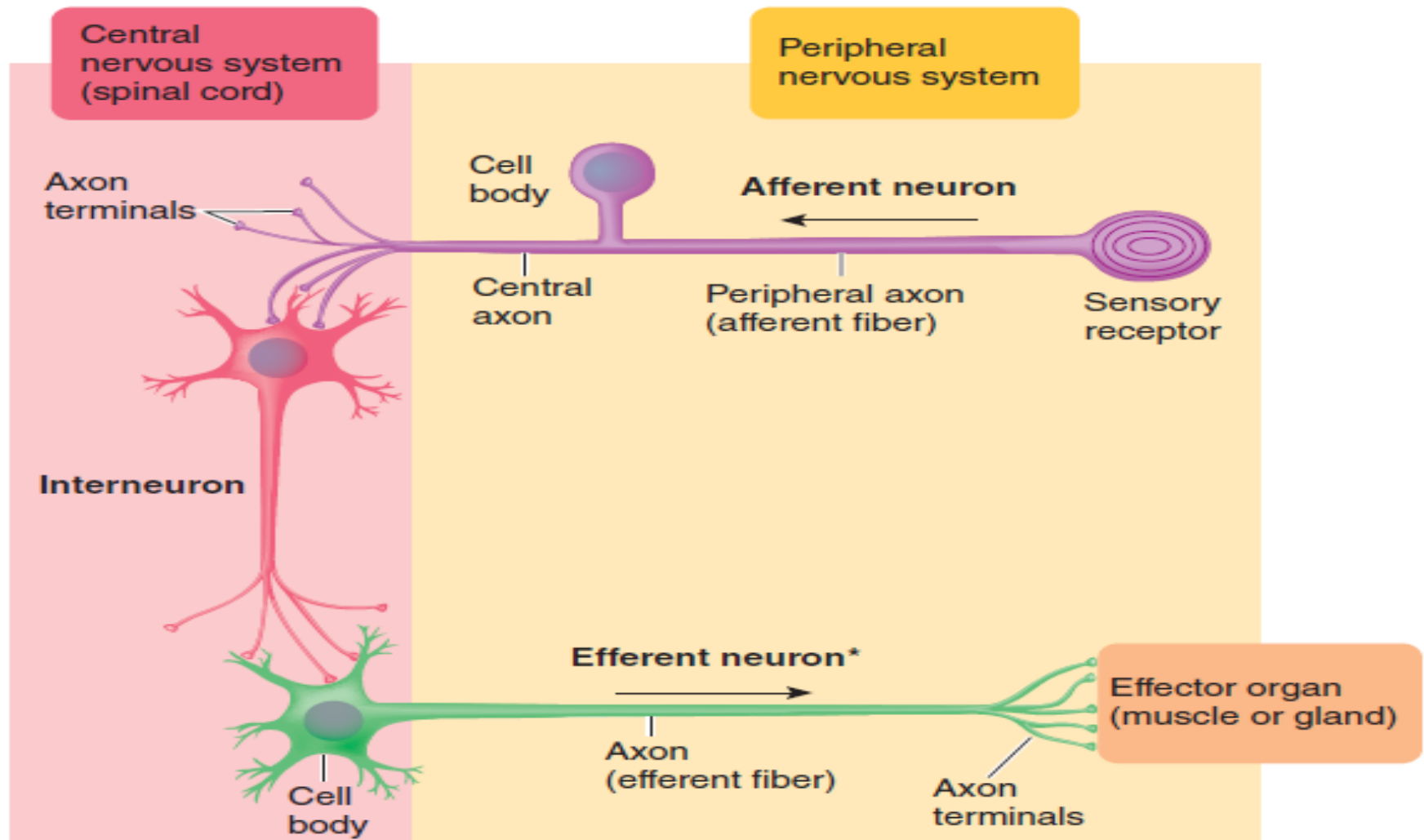
3. Interneurons

- Function as integrators and signal changers.
- Integrate groups of afferent and efferent neurons into reflex circuits.
- Lie entirely within CNS.
- Account for 99% of all neurons (A: E: I, 1:10:200,000)

Classification of Neurons



Classification of Neurons



Classification of Neurons

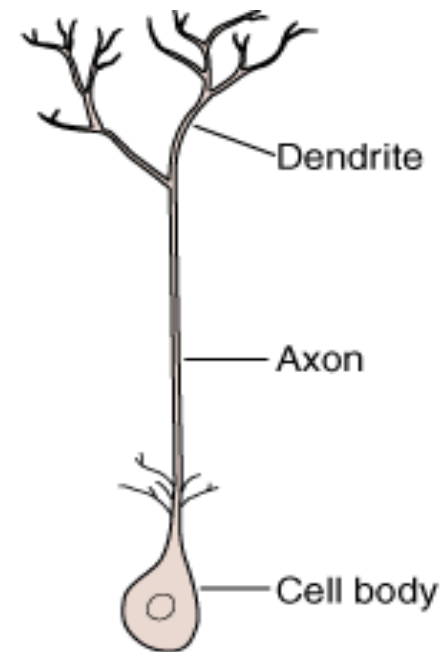
B. On the basis of number of processes that originate from the soma

1. Unipolar Neurons

- Have a single primary process
- Invertebrate nervous system
- In vertebrate: Autonomic nervous system

2. Pseudo-unipolar Neurons

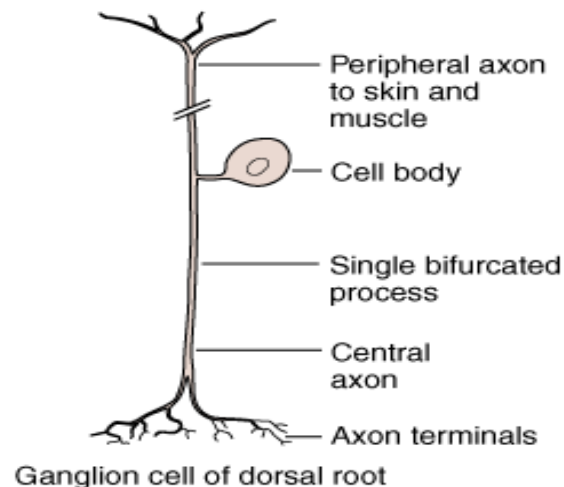
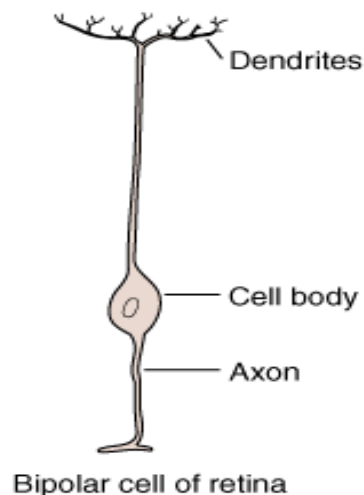
- Mechanoreceptors: touch, pressure, stretch and pain...
- (Posterior root or cranial nerve ganglion)



Classification of Neurons

3. Bipolar Neurons

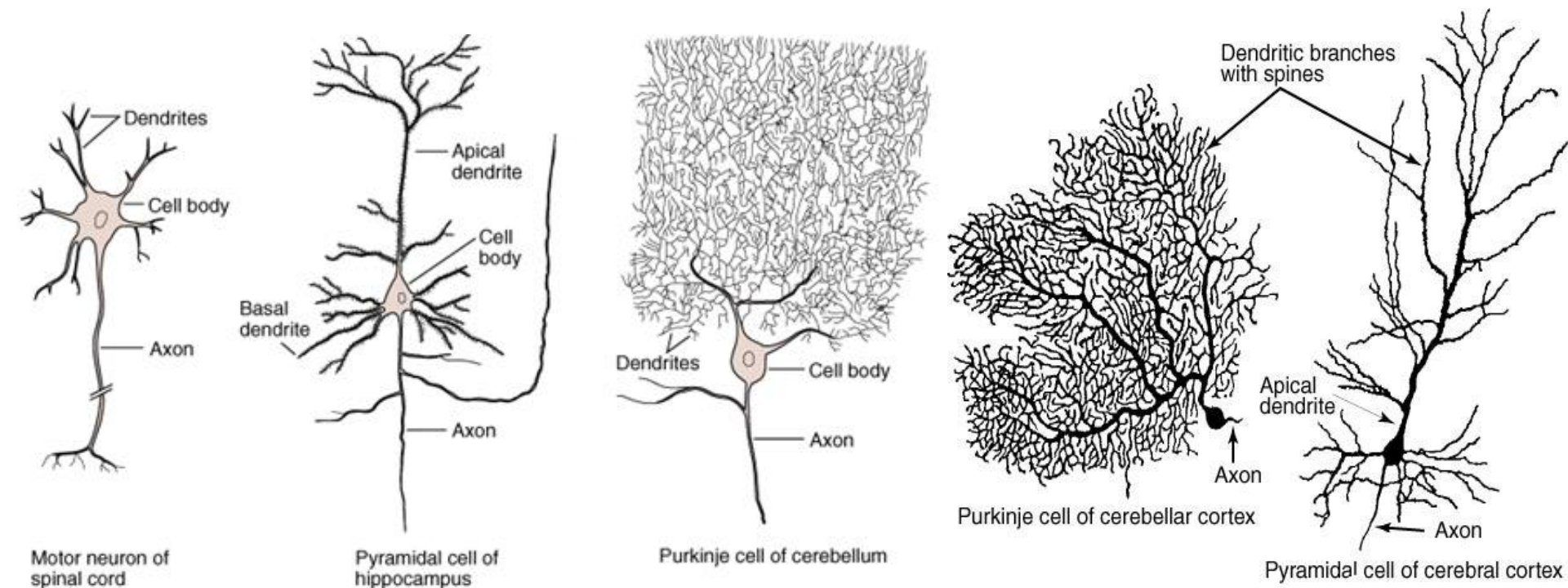
- Two processes (axon + dendrites)
 - Dendrite: conveys information from the periphery of the body.
 - Axon: carries information toward the CNS
- Many of sensory cells: retina, auditory, vestibular, olfactory



Classification of Neurons

4. Multipolar Neurons

- Predominate in the nervous system of vertebrates
- Single axon and many dendrites



Classification of Neurons

C. On the basis of electrical activity

1. Silent Neurons

- Steady unchanging RMP in the absence of external stimulation.

2. Pacing Neurons/Beating Neurons

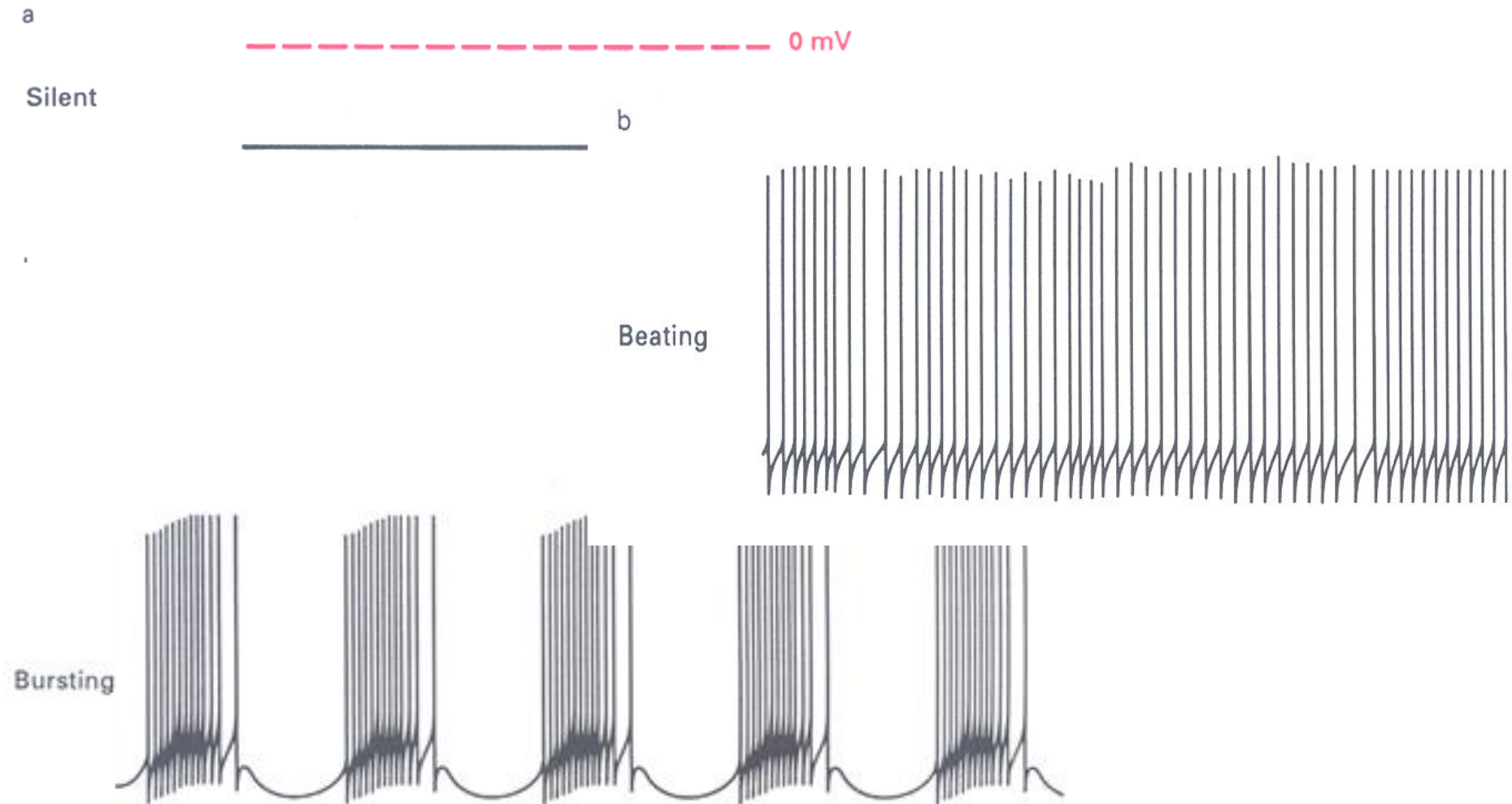
- Fire repetitively at a constant frequency (*do not require continual synaptic activation or other external stimuli*)
- External stimulation can change the firing rate of the cell or inhibit it altogether

Classification of Neurons

3. Bursting Neurons

- Fire spontaneously in the absence of external stimulation
- Do not fire at fixed regular intervals
- Generate regular bursts of action potentials that are separated by hyperpolarization of the membrane
- Significance:
 - Generate rhythmic behaviors (*breathing, walking, swimming...*)
 - Secrete neurohormones (OXT, AVP)

Classification of Neurons



Classification of Neurons

- On the type of NT they synthesize and release
 - Glutamatergic
 - Cholinergic
 - Adrenergic... etc.,
- On the basis of their shape
 - Pyramidal
 - Granular
 - Mitral...
- On the basis of their location
 - Cortical neurons
 - Spinal neurons etc.
- On length of axons
 - Projecting/Relay/Golgi type I neurons
 - Intrinsic/interneurons/Golgi type II
- Others (Nature of dendrites, myelination, diameter of axon & velocity of AP... etc.,)

Glial Cells

- Neuroglia (Glia → Nerve glue)
- Supportive matrix.
- 10^{11} glial cells (1:1)
- NOT directly involved in signal processing

Peripheral Glial Cells

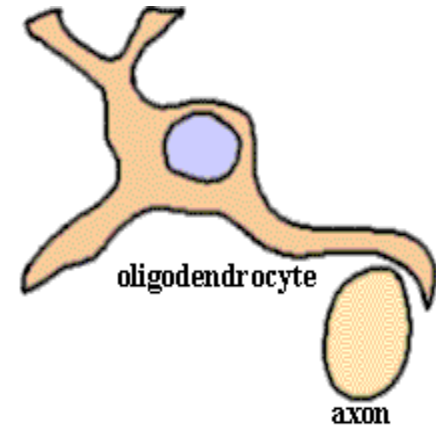
A. Schwann cells

- Produce **Myelin sheath** in the PNS
- Myelin: Lipids (Cholesterol, Phospholipids, 70%)
Proteins (30%)
- Envelope ONLY ONE AXON (Insulation→ speeds transmission of signals along nerves)

B. Satellite cells

- Encapsulate dorsal root and cranial nerve ganglia

Myelination in the CNS →

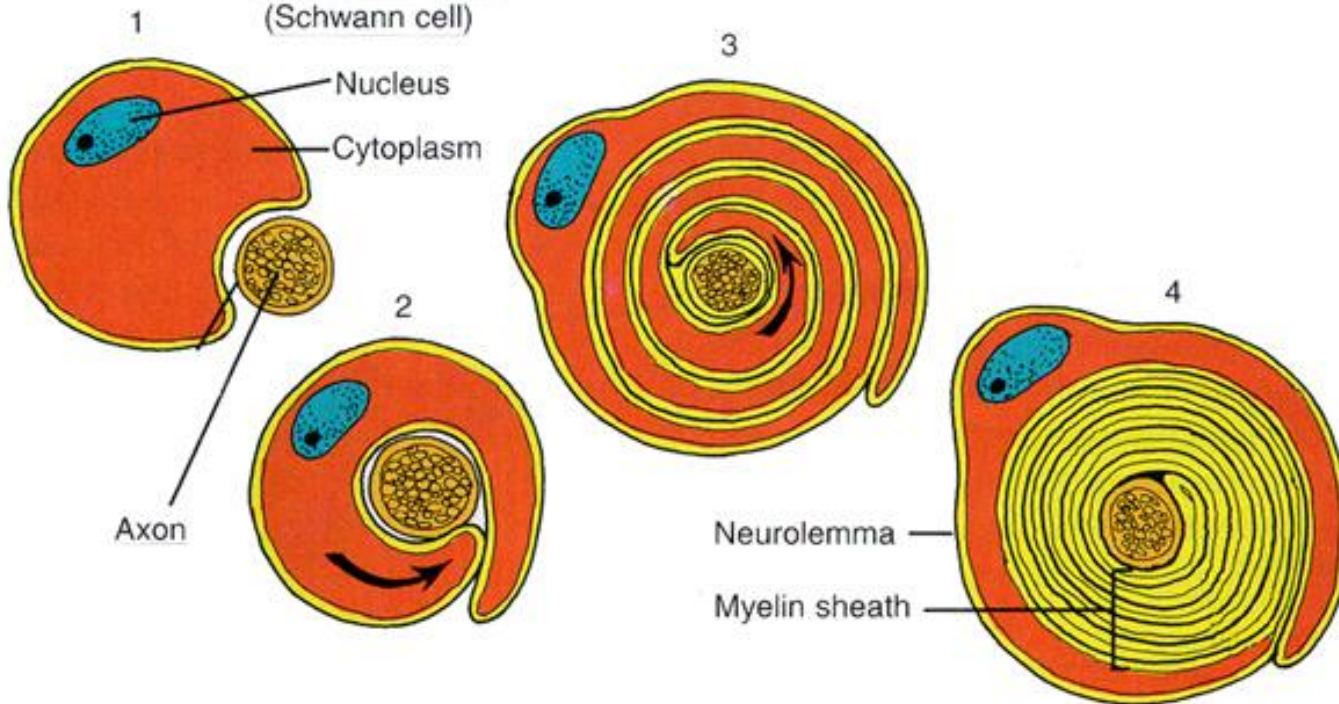


The Process of Myelination

Myelination in the PNS



Neurolemmocyte
(Schwann cell)



Central Glial Cells

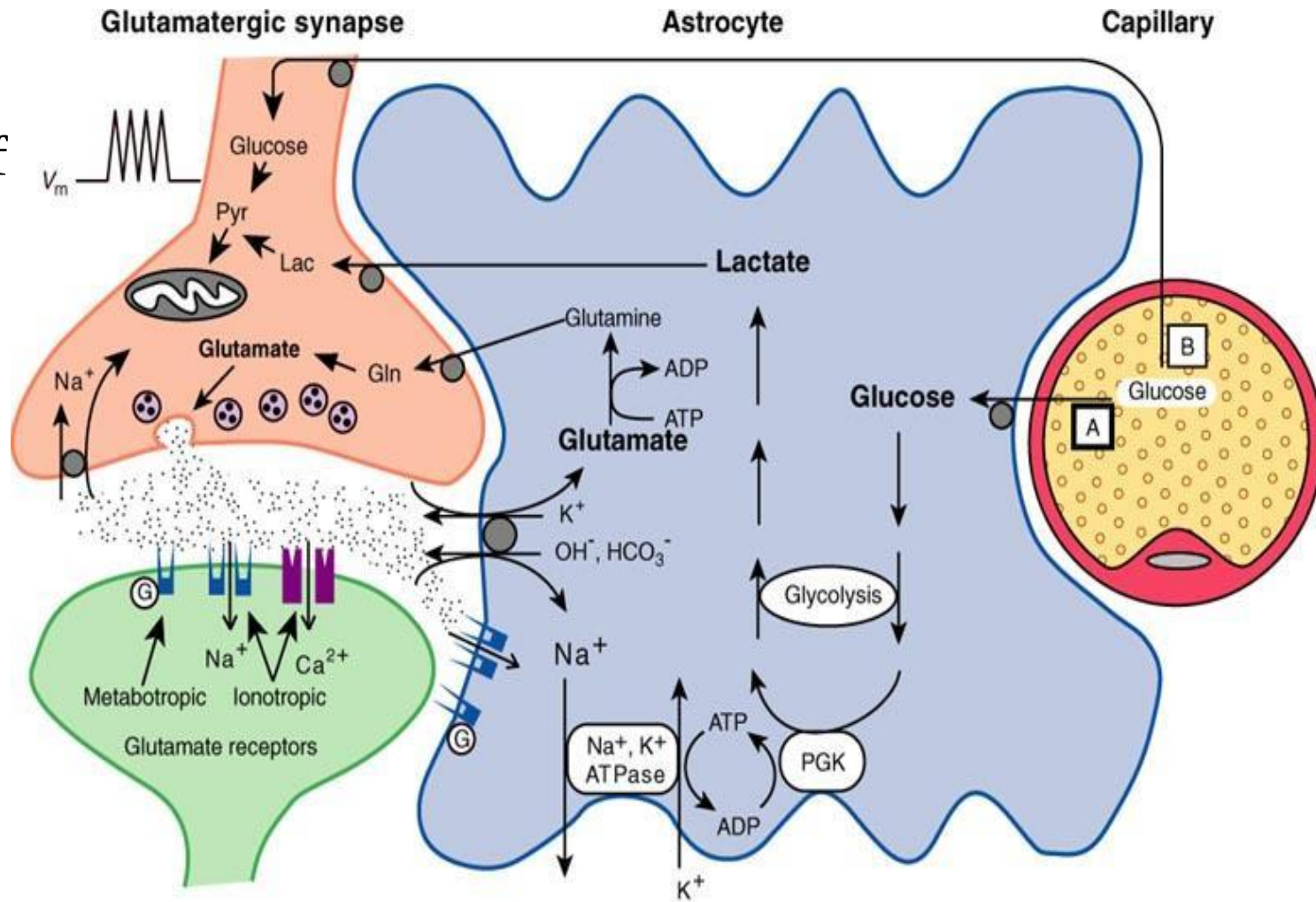
A. Astrocytes

- Star-shaped, abundant, and versatile
- Provide firmness + structural support to the brain
- Nutritive support
- Spatial buffering effect of K^+
- Re-uptake of neurotransmitters
- Induce formation of the Blood-Brain Barrier/BBB
- Detoxification
- Produce growth factors (BDNF, GDNF, PDGF, IGF-1...)

Central Glial Cells

A. Astrocytes

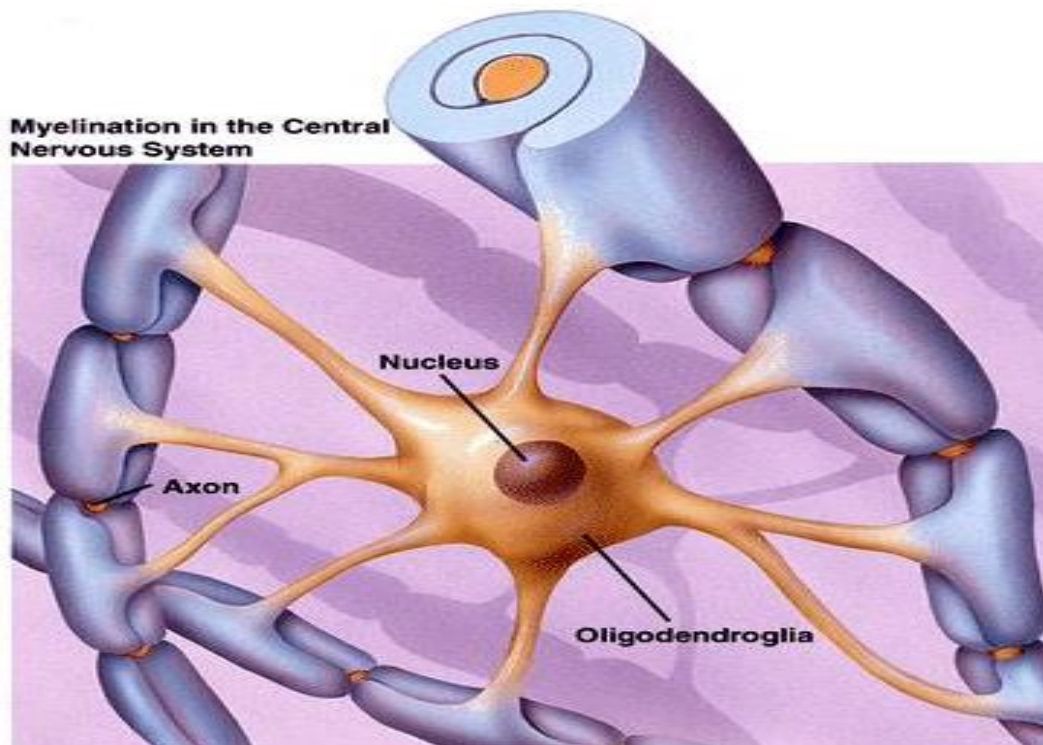
- Formation of synapses



Central Glial Cells

B. Oligodendrocytes

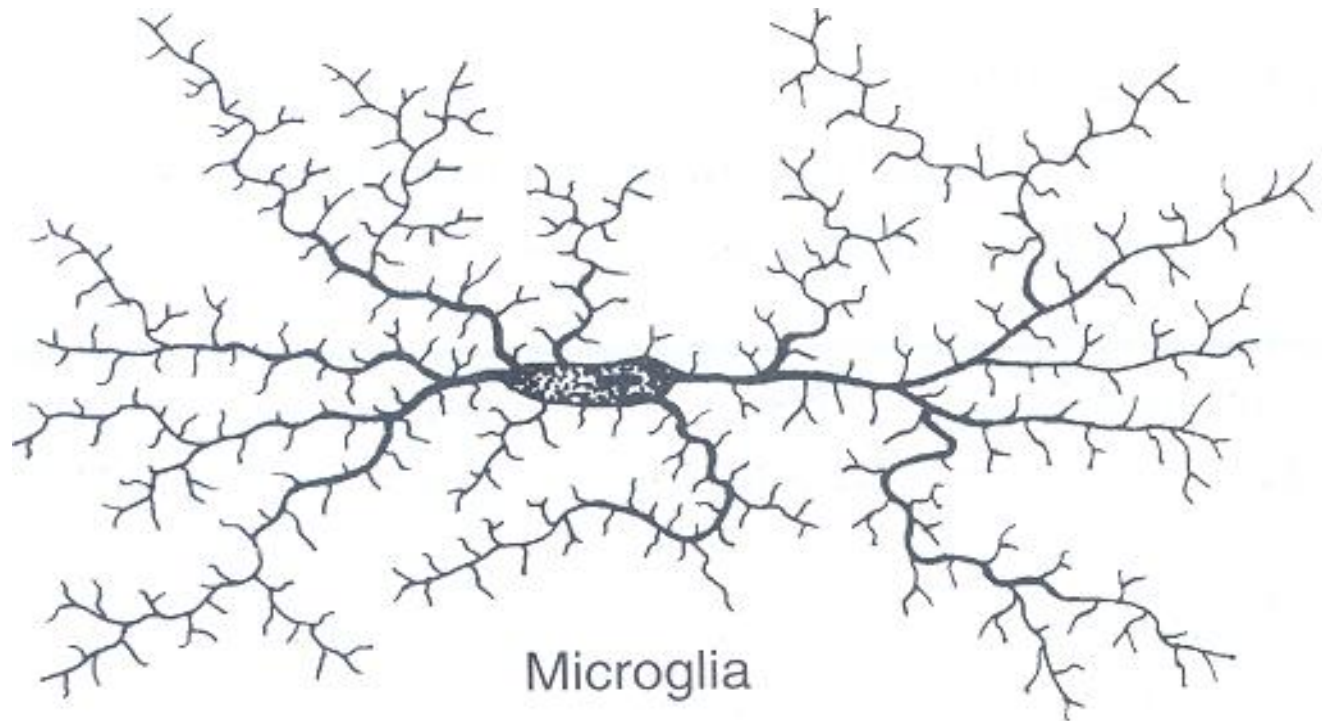
- Form **myelin sheath** in the CNS (X=40-60 axons)
- Insulate groups of neurons from each other.



Central Glial Cells

C. Microglia

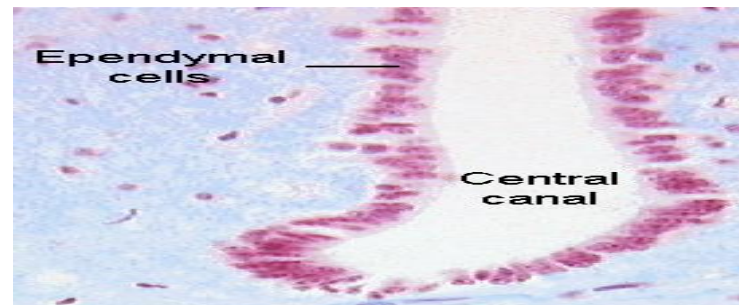
- Immune cells of the CNS
- Scavenging: removing debris after infection, injury or neuronal death



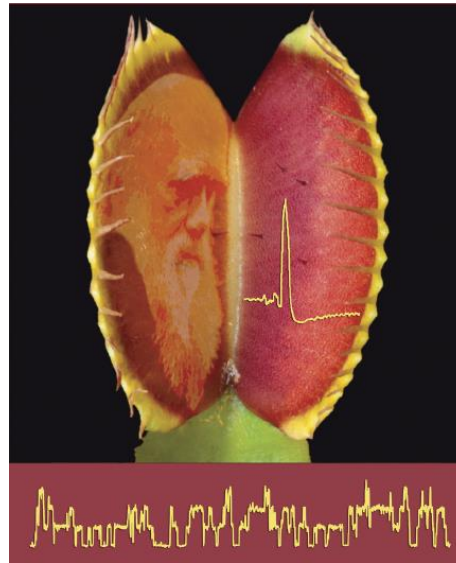
Central Glial Cells

D. Ependymal Cells

- Low columnar epithelial cells that line the ventricles of the brain and the central canal of the spinal cord
- Production of CSF
- Some are ciliated which facilitates the movement of CSF
- Form the epithelium that separates CNS from CSF
- Assist in the circulation of CSF within the brain and central canal of the spinal cord
- Absorptive role



Basic Principles of Electrophysiology



Membrane Potential

Basic principles

- Current flow:
 - Cations and anions carry electrical current
 - Current flow → direction of net movement of positive charge
 - Cations move in the same direction as the current
 - Anions move in the opposite direction as the current

Basic principles

- Inward current:
 - Flow of positively charged ions (e.g. Na^+) into the cell
- Outward current:
 - Flow of Positively charged ions (e.g. K^+) out of the cell; or
 - Flow of Negatively charged ions (e.g., Cl^-) into the cell
- Leakage current:
 - Current due to the flow of ions through the non-gated ion channels

Membrane Potential

- Electrical energy difference between the inside and outside of the cell

$$E_m = V_{in} - V_{out}$$

where:

$E_m \rightarrow$ Membrane potential (**mV**)

$V_{in} \rightarrow$ Potential on the inside of the cell

$V_{out} \rightarrow$ Potential on the outside of the cell

Membrane Potential...

- All cells have membrane potentials
- Membrane potential $\propto$ charge separation across the membrane.
- Range of E_m : **-20 mV to -90mV (organism, cell type)**
- Any change of membrane's permeability to ions causes a change in E_m

Resting Membrane Potential

- Steady transmembrane potential of a cell that did not produce an electrical signal
- No net flow of ions across the plasma membrane (*No net inward current*)
- Magnitude is constant for individual cell types:
 - Nerve, cardiac and skeletal muscle → -55 to -90mV
 - Smooth muscle → -55 to -30mV
- RMP is necessary for the cell to fire an action potential
- RMP is nearly equals to that of E_K

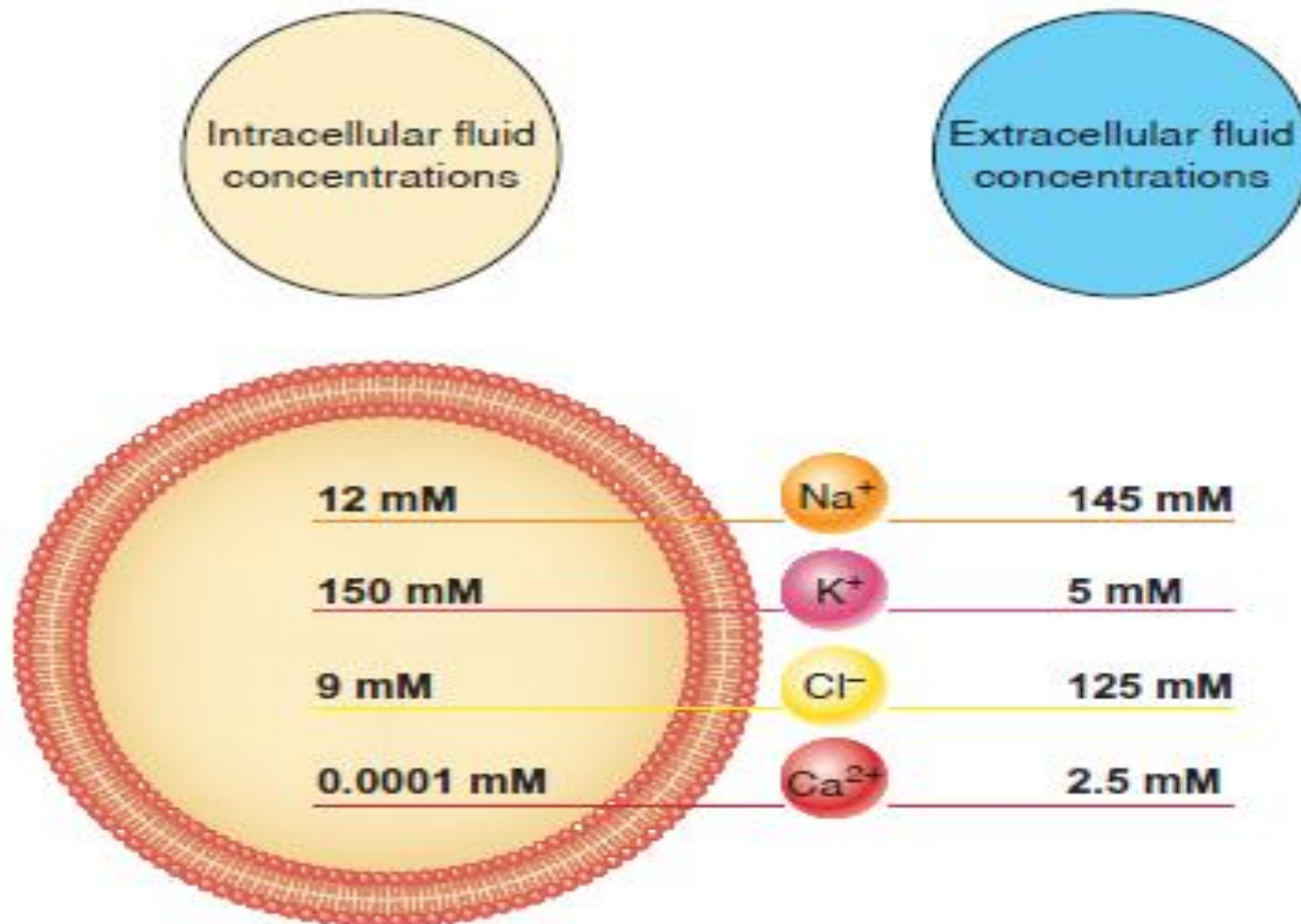
Membrane Potential: Determinants

1. Passive determinants
2. Active determinant

1. Passive Determinants:

- a. Biochemical nature of the plasma membrane
- b. Uneven distribution of ions across the membrane

Distribution of ions across the membrane



Concentration of ions

	<u>Inside</u>	<u>Outside</u>	<u>conc. gradient</u>	<u>Electrostatic pressure</u>
Sodium (Na^+)	12mM	145mM	in to cell	in to cell
Potassium (K^+)	150mM	5mM	out of cell	in to cell
Chloride (Cl^-)	9mM	125mM	in to cell	out of cell
Calcium (Ca^{2+})	10^{-4} mM	2.5mM	in to cell	in to cell
Organic anions:	Fixed anions (Proteins, nucleotides, polyphosphates...)			

Membrane Potential: Determinants

c. Diffusional force

- Potential difference between inside and outside of the cell

• Nernst Equation/Nernst Potential

- Ion concentrations and equilibrium potentials:
 - Conc. of the ion inside and outside
 - Temperature of the solution
 - The valence of the ion
 - The amount of work required to separate a given quantity of charge

$$E_x \text{ (mV)} = \frac{RT}{zF} \ln \frac{[X]_o}{[X]_i}$$

Where:

R = gas constant = $8.315 \text{ J K}^{-1} \text{ mol}^{-1}$

T = $T_{\text{Kelvin}} = 273.15 + T_{\text{Celsius}}$

F = Faraday's constant = $96,485 \text{ C mol}^{-1}$

z = Valence of the ion (+1, -1, +2)

$[X]_o$ = Concentration of the ion outside the cell

$[X]_i$ = Concentration of the ion inside the cell

$$E_x = \pm 61 \log_{10} \frac{[X]_o}{[X]_i}$$

NB. Nernst Equation predicts Equilibrium Potential of ions (E_{ion}).

Goldman-Hodgkin-Katz Equation/The Constant field Equation

- Polarity
- Permeability
- Concentration

$$E_m = \frac{RT}{F} \ln \frac{P_K [K^+]_o + P_{Na} [Na^+]_o + P_{Cl} [Cl^-]_i}{P_K [K^+]_i + P_{Na} [Na^+]_i + P_{Cl} [Cl^-]_o}$$

(Permeability = $P_K : P_{Na} : P_{Cl}$, 1.00: 0.04: 0.45)

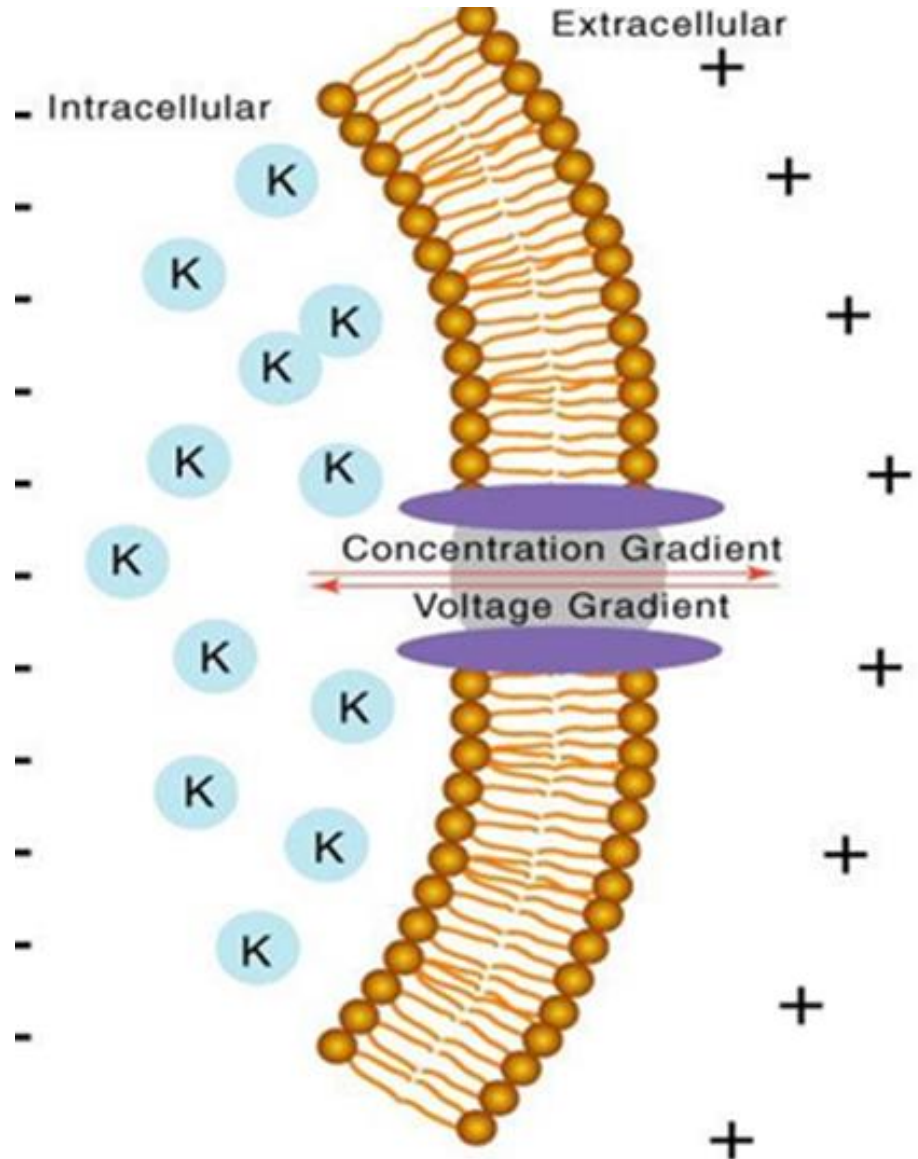
Goldman-Hodgkin-Katz Equation

1. Contribution of any ion to the E_m is determined by the **membrane's permeability** to that particular ion.
2. The higher the permeability of the membrane to one ion relative to the others, the more that ion will contribute to the E_m .
3. Contribution of the **impermeable ion** to the E_m will be zero.

NB. GHK → predicts Resting Membrane potential of a cell.

c. Diffusional force

- Net driving force on ion X:
 - Concentration gradient
 - Electrical gradient



d. Leaky channels (non-gated channels)

- E_m is directly proportional to the concentration gradient of the most permeable ion:
 - At rest $E_m \sim E_{K^+}$

2. Active Determinant

❖ Na^+ - K^+ -ATPase (Na^+ - K^+ pump)

I. Features:

- a. A carrier molecule uses the membrane-bound ATPase
- b. Primary active transport process (consumes ATP, pumps against electrochemical gradient)
- c. Operates as antiporter (coupled transporter):
 - Pump **3Na^+** out of the cell and **2K^+** in to the cell
(*Electrogenic pump*)

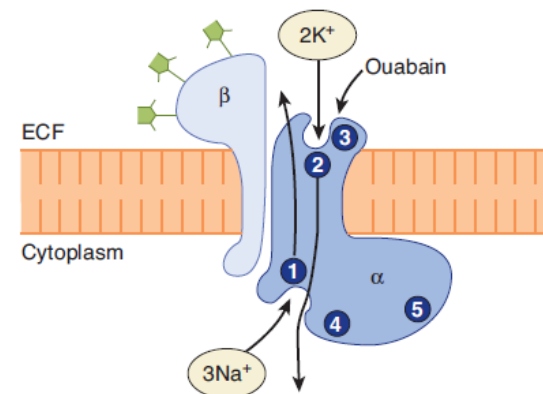
❖ Na^+ - K^+ -ATPase (Na^+ - K^+ pump)...

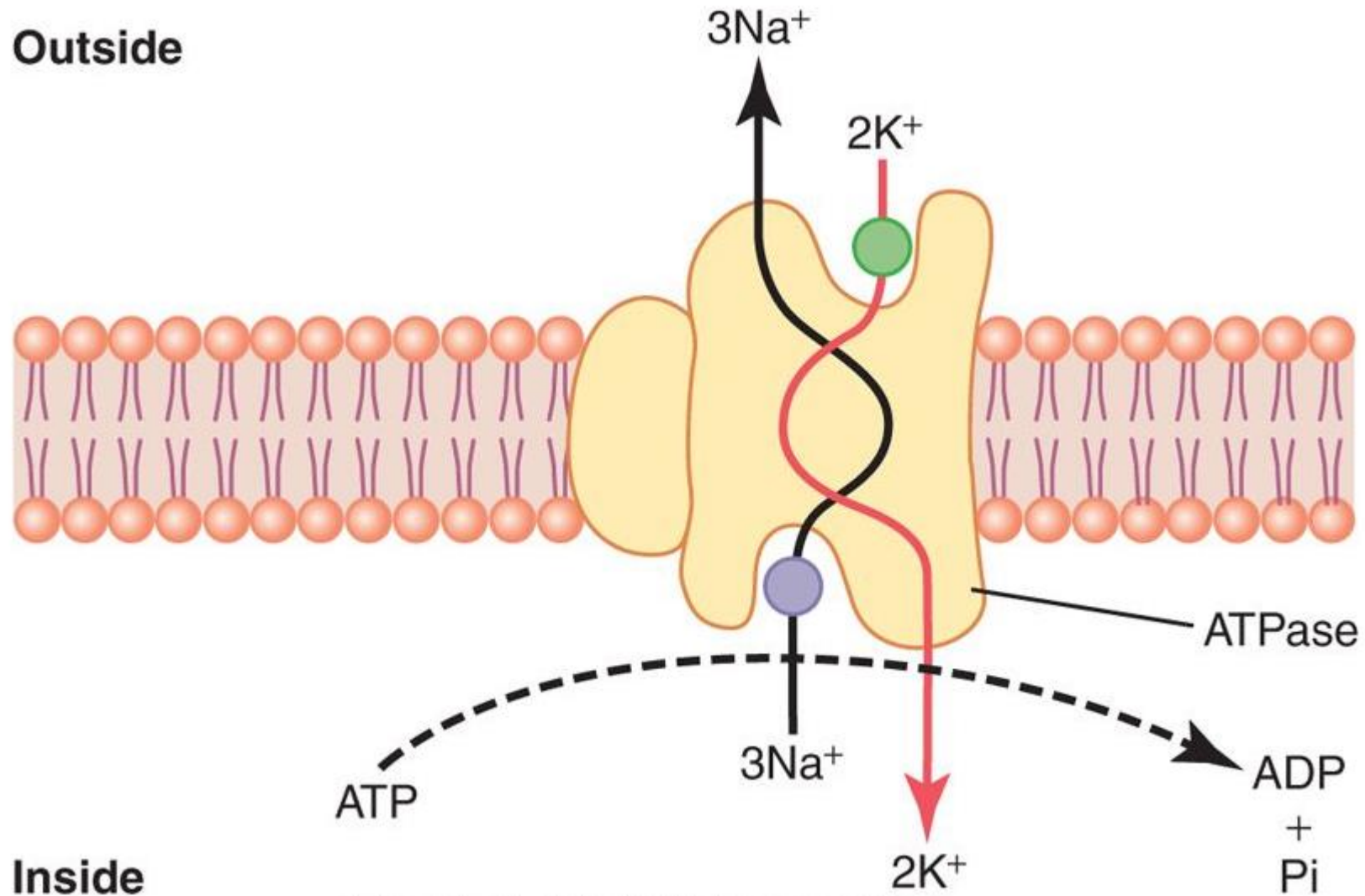
II. Location:

- Plasma membrane of virtually all animal cells:
 - 200 Na^+ - K^+ -ATPase/RBC
 - 35,000 Na^+ - K^+ - ATPase /WBC

III. Function

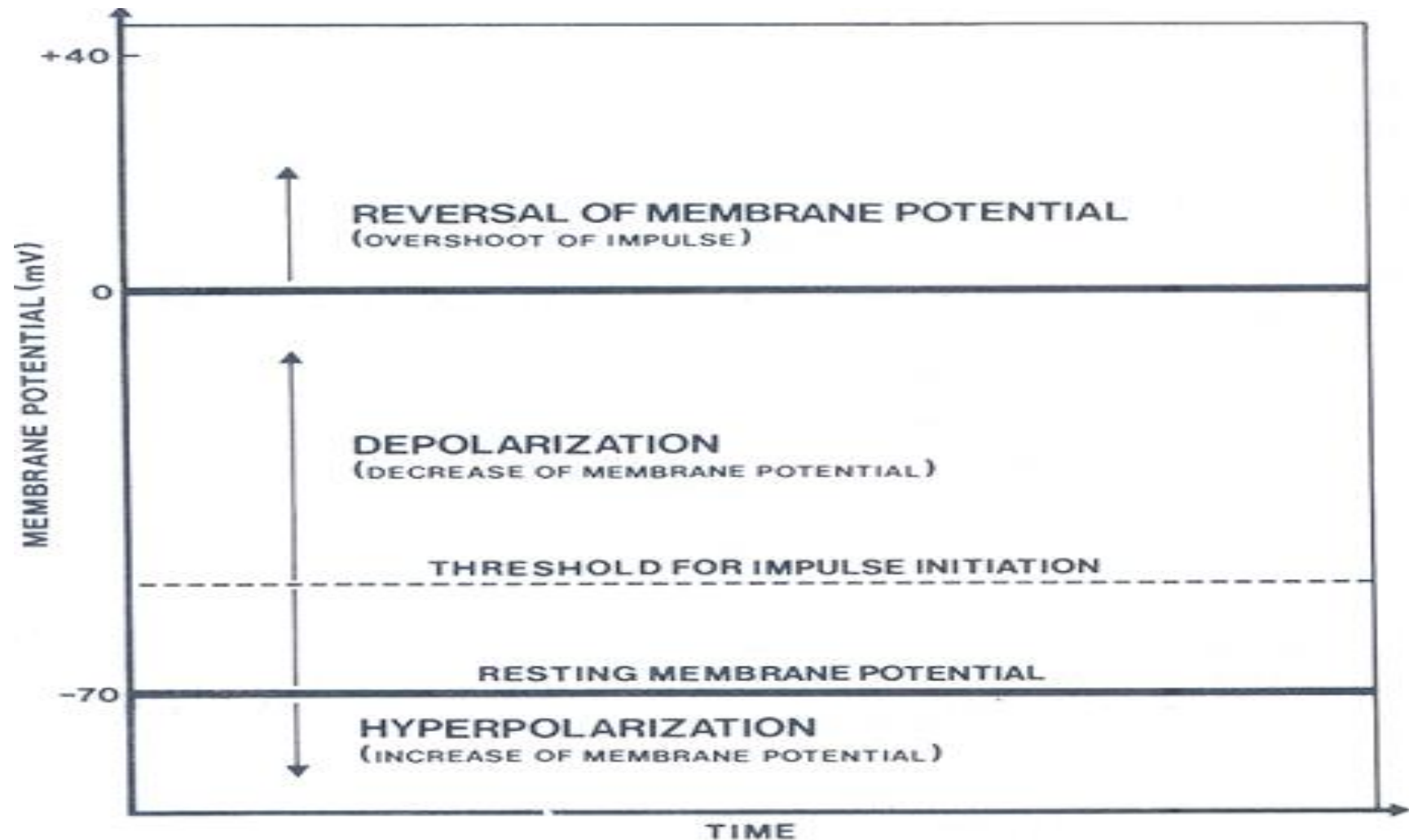
- Maintaining the gradient of Na^+ and K^+ across the cell membrane:
 - Cell volume
 - Excitability

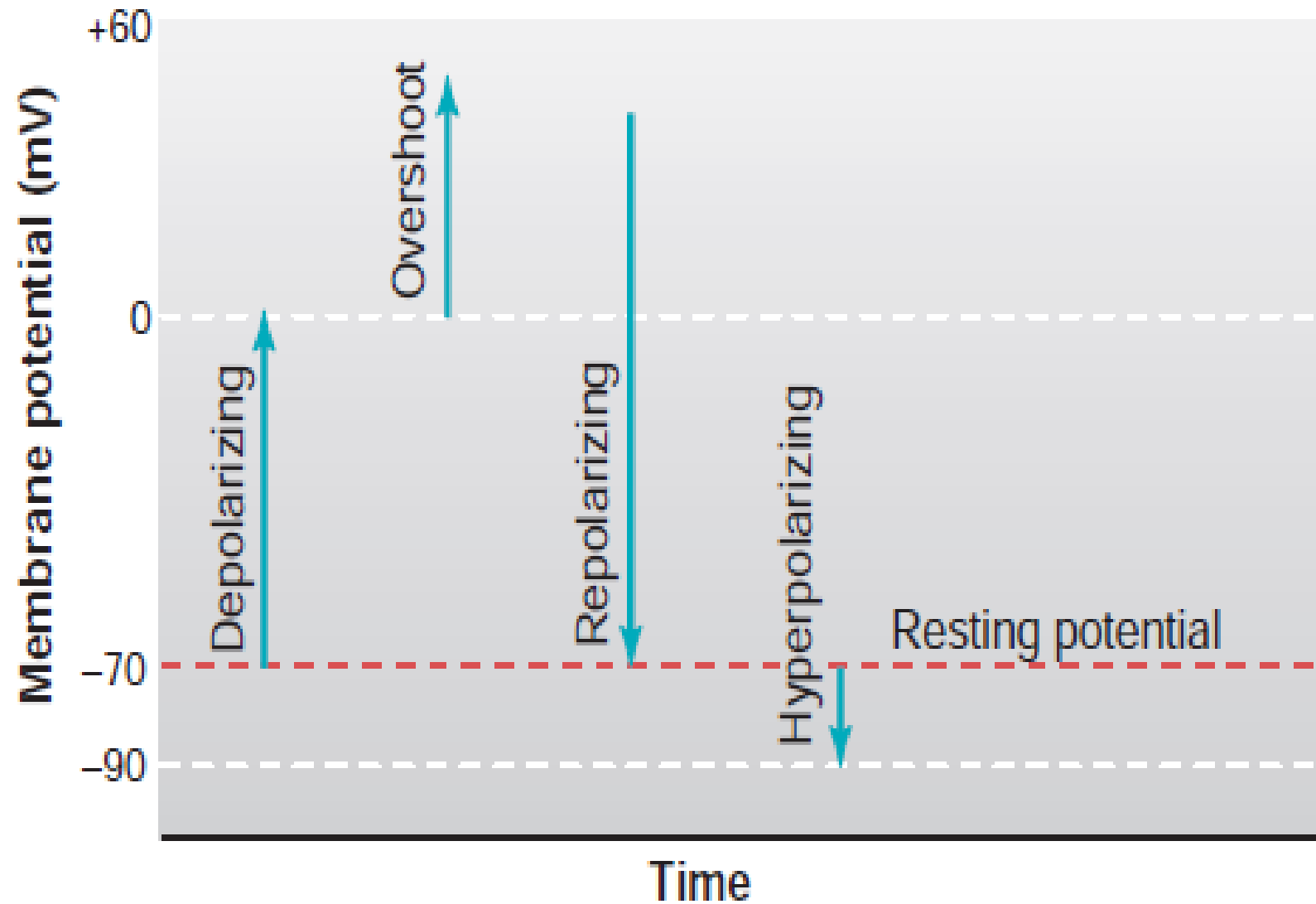


**Na⁺-K⁺-ATPase**

Neural Signaling

$\Delta E_m \rightarrow$ Basis of signaling in the NS





Neural Signaling

Types: 1. Graded potentials
2. Action potentials

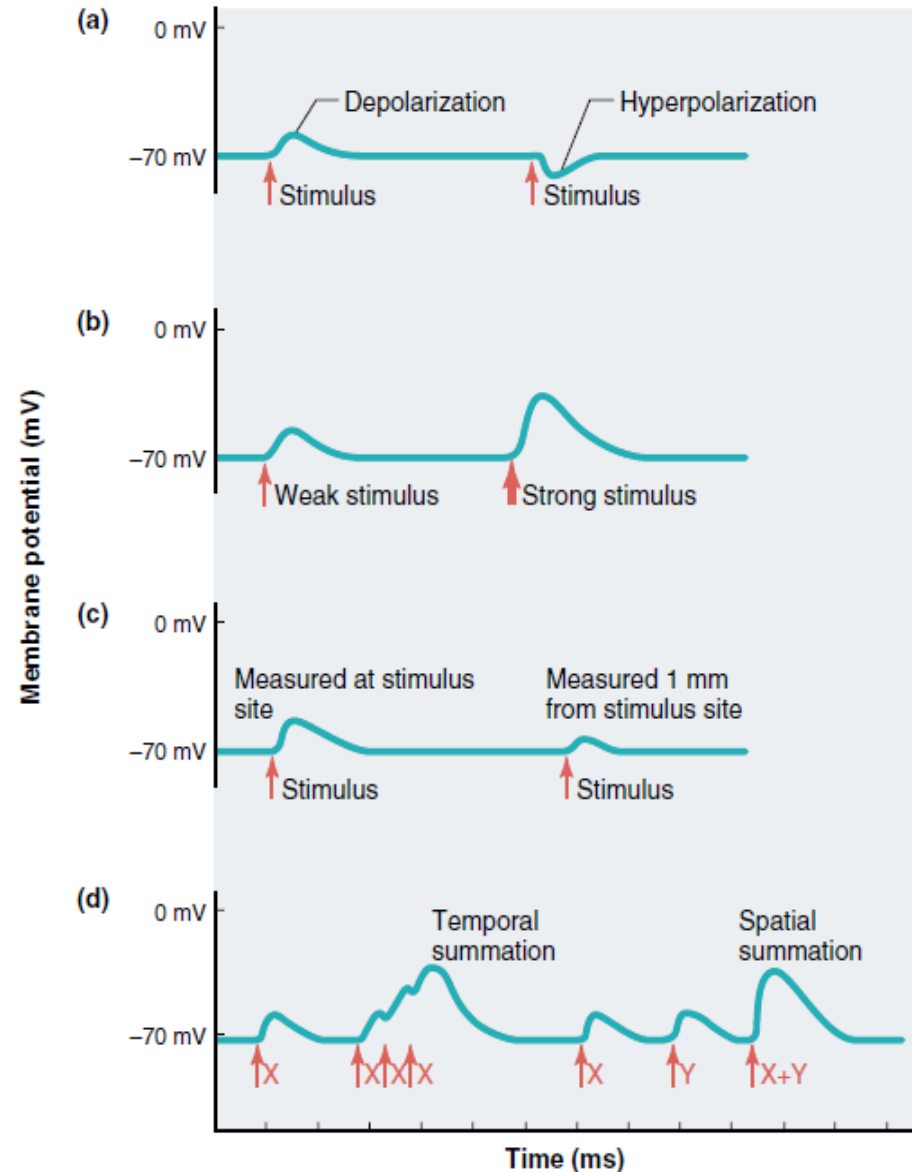
1. Graded Potentials

a. Def.:~ local changes in E_m in either depolarizing or hyperpolarizing direction

1. Graded Potentials

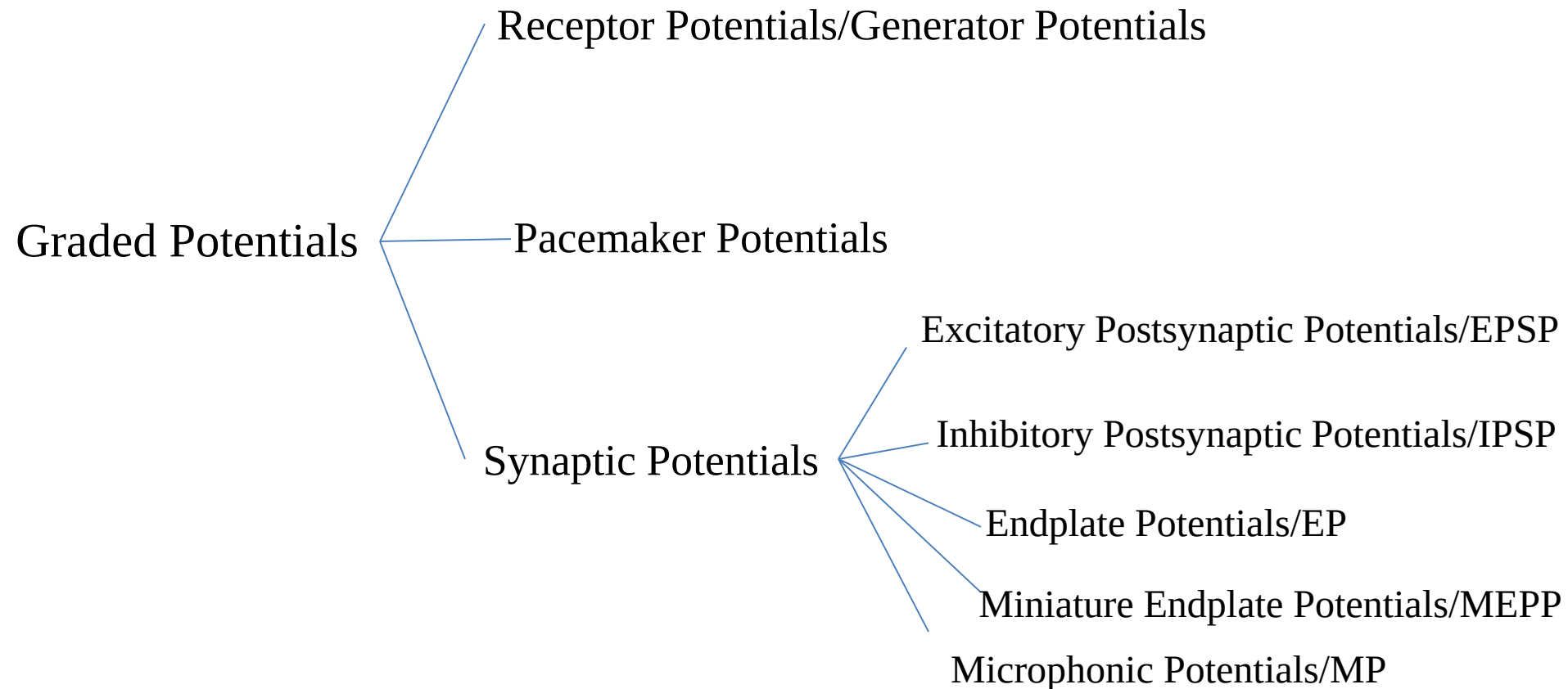
b. Features:

1. Depolarizing or hyperpolarizing
2. Variable in amplitude and duration
3. Conducted decrementally
4. Can be summed
5. Has no threshold
6. Has no refractory period
7. Propagation is passive
8. Mediated by a receptor



1. Graded Potentials

c. Types:



d. Mechanism of signaling:

Stimulus (physical, mechanical, chemical, electrical...)



Sensory receptors



Transform stimulus energy/Transduction



Opening of Ion channels



Inward flow of current (Na^+ ↓, K^+ ↑)



Depolarization → ΔE_m



Receptor potential/Generator potential



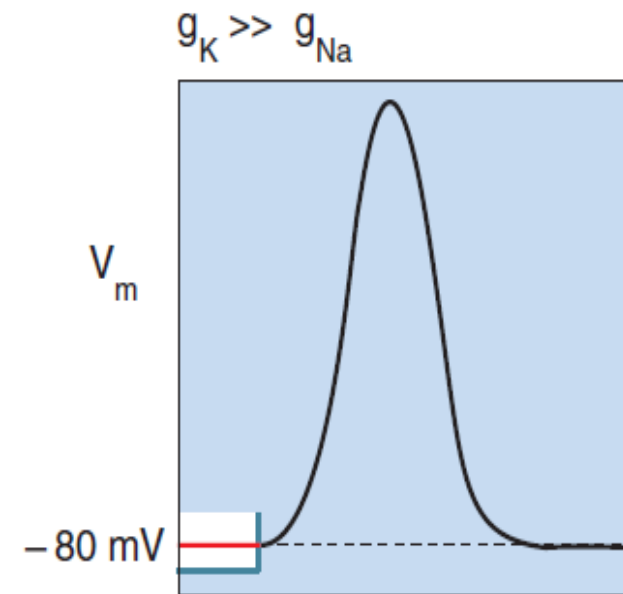
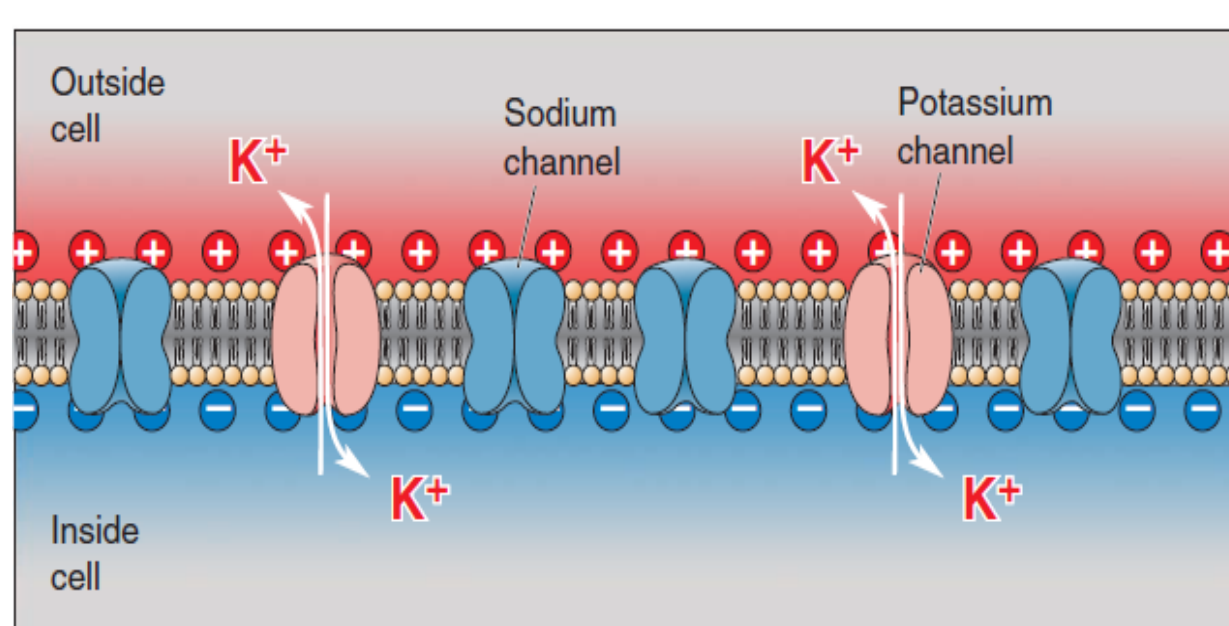
Action Potential



CNS ... → RESPONSE.

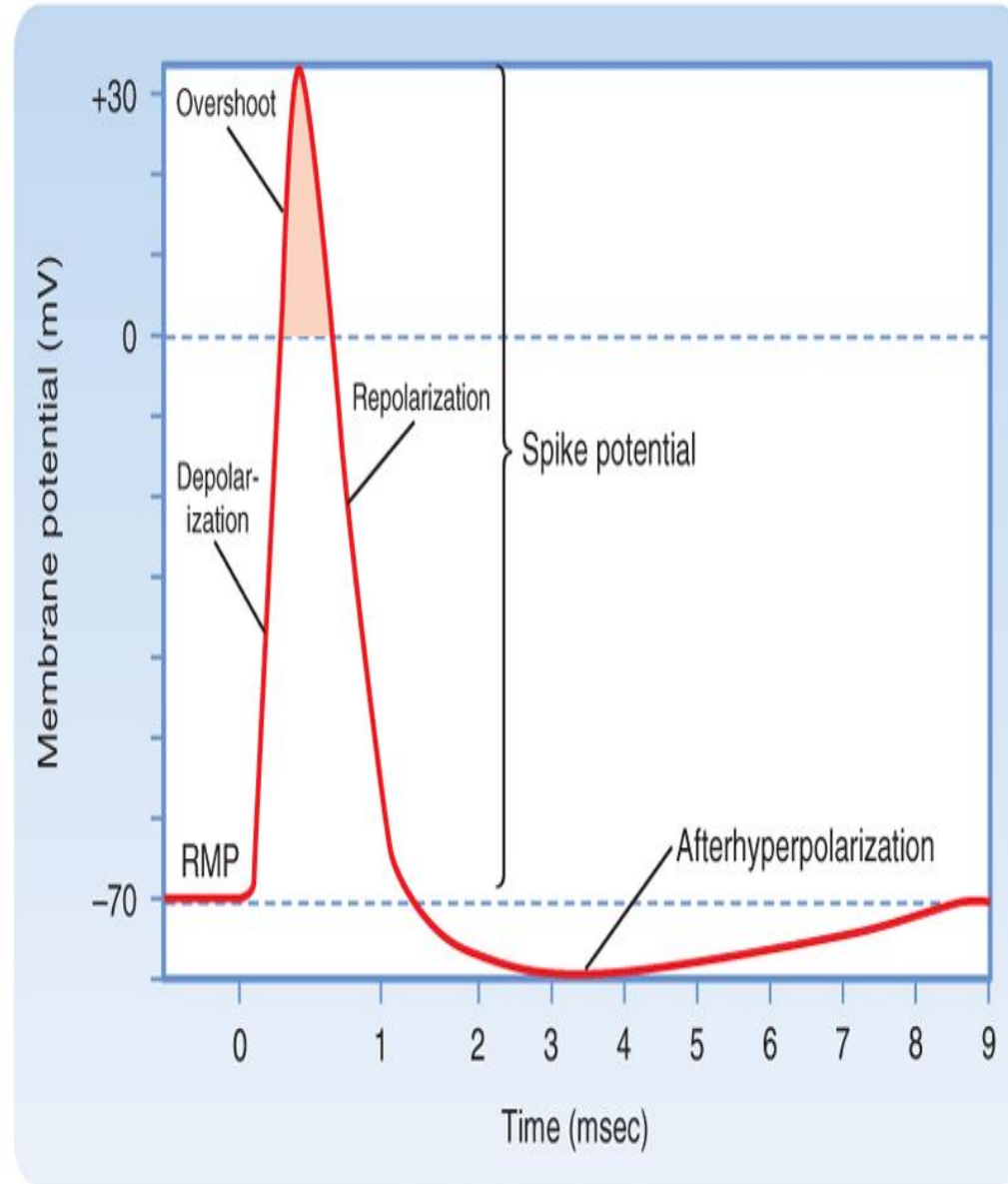
2. Action Potentials

- a. *Def:* ~ **rapid, transient, self-propagating** electrical excitation in the plasma membrane of excitable cells.
- b. *Genesis/Initiation:* electrical perturbation $\rightarrow \Delta E_m \rightarrow$ AP.
- Sequential opening of voltage-gated channels of Na^+ and K^+

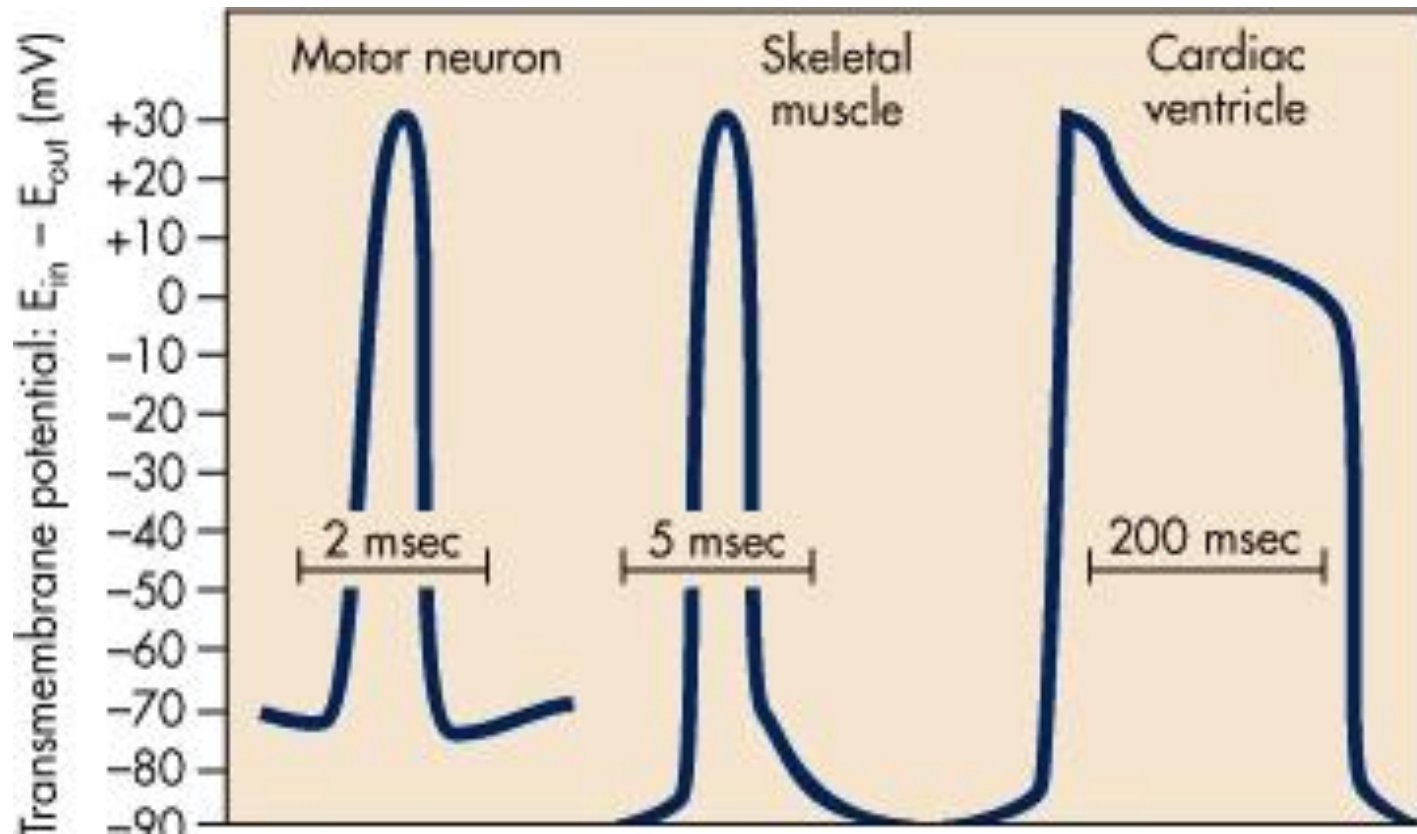


• AP Features

1. *All-or-none* phenomenon
 - Always full sized
 - It will not get lost along the way.
 - Rapid and reliable information
2. Has threshold
3. Amplitude and duration is κ
4. Always depolarizing
5. Has refractory period
6. Non-decremental
7. Propagation is active
8. Mediated by VG channels



- SAME SHAPE, AMPLITUDE + DURATION in same excitable cells.

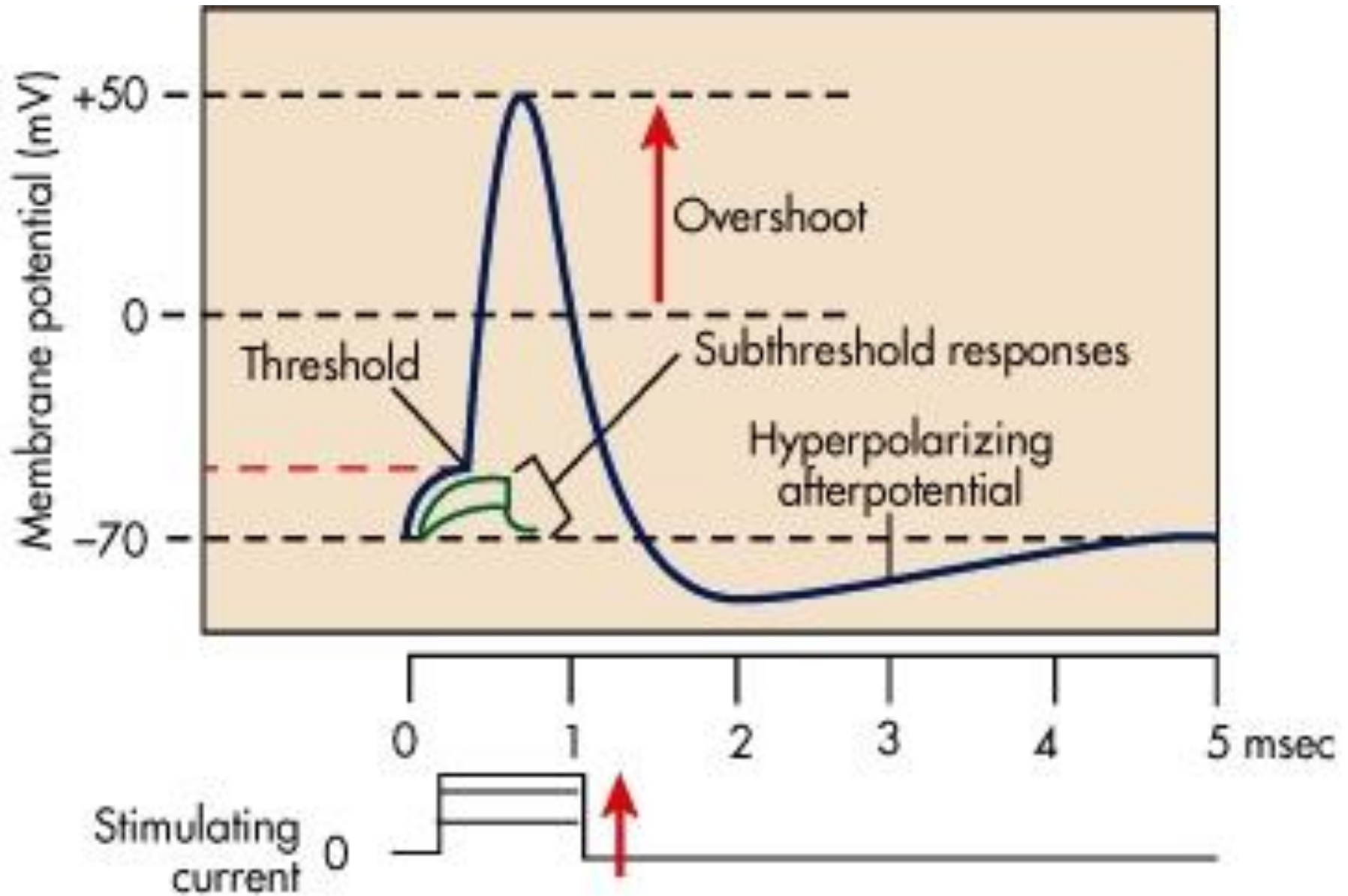


- Shape, amplitude and duration of AP is different in different excitable cells

d. Phases and Ionic Basis of Action Potential

1. Threshold Potential

- i. Minimum value of E_m at which an action potential will occur
- ii. E_m is reduced to a critical level $\rightarrow$ excitable cells undergo rapid depolarization
 - Initiated by rapid opening of fast Na^+ channels.
- iii. AP occurs only when the NET movement of positive charge is inward (**$g_{\text{Na}}^+ > g_{\text{K}}^+$ or Na^+ influx $>$ K^+ efflux**)

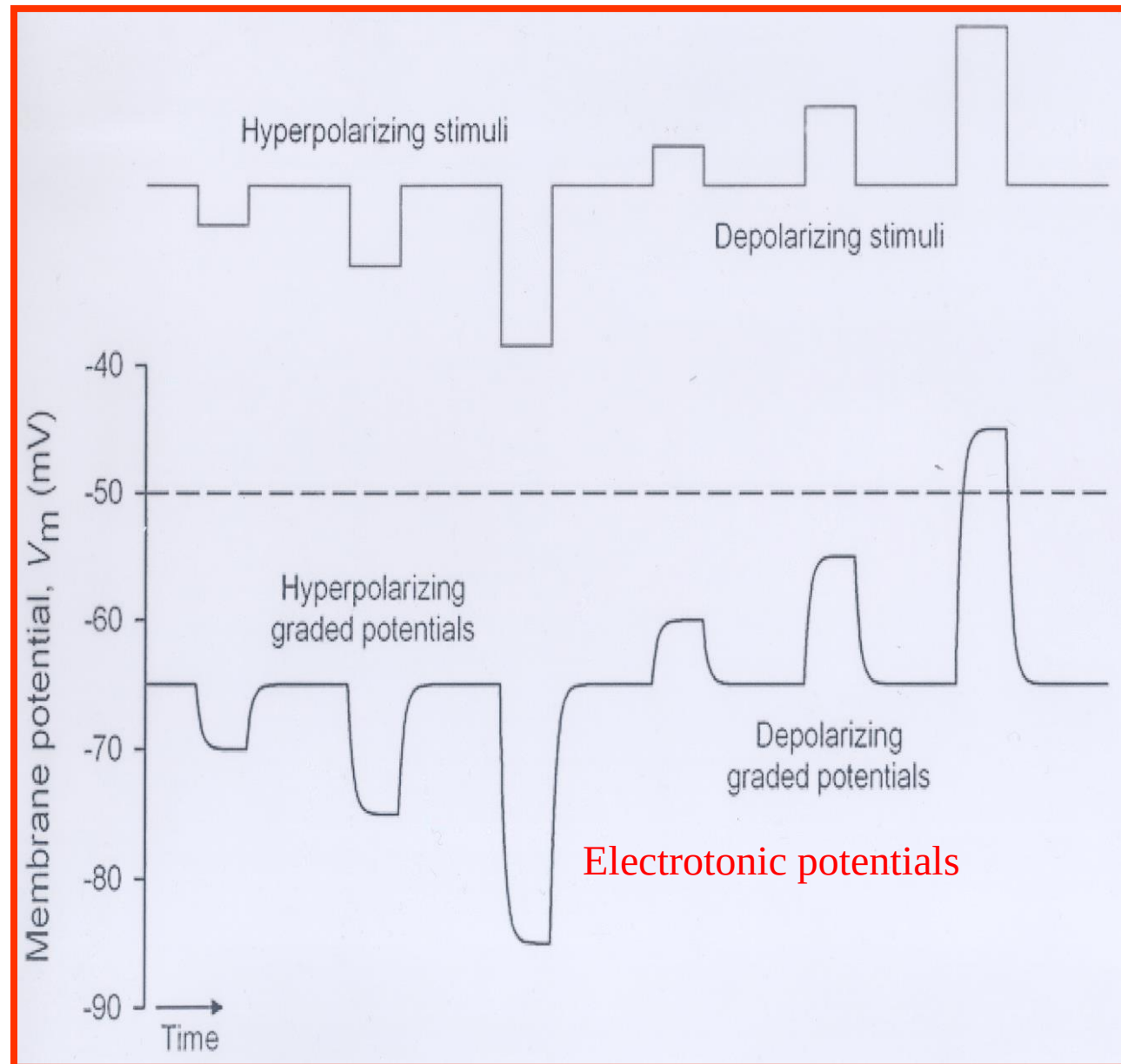


Threshold Potential

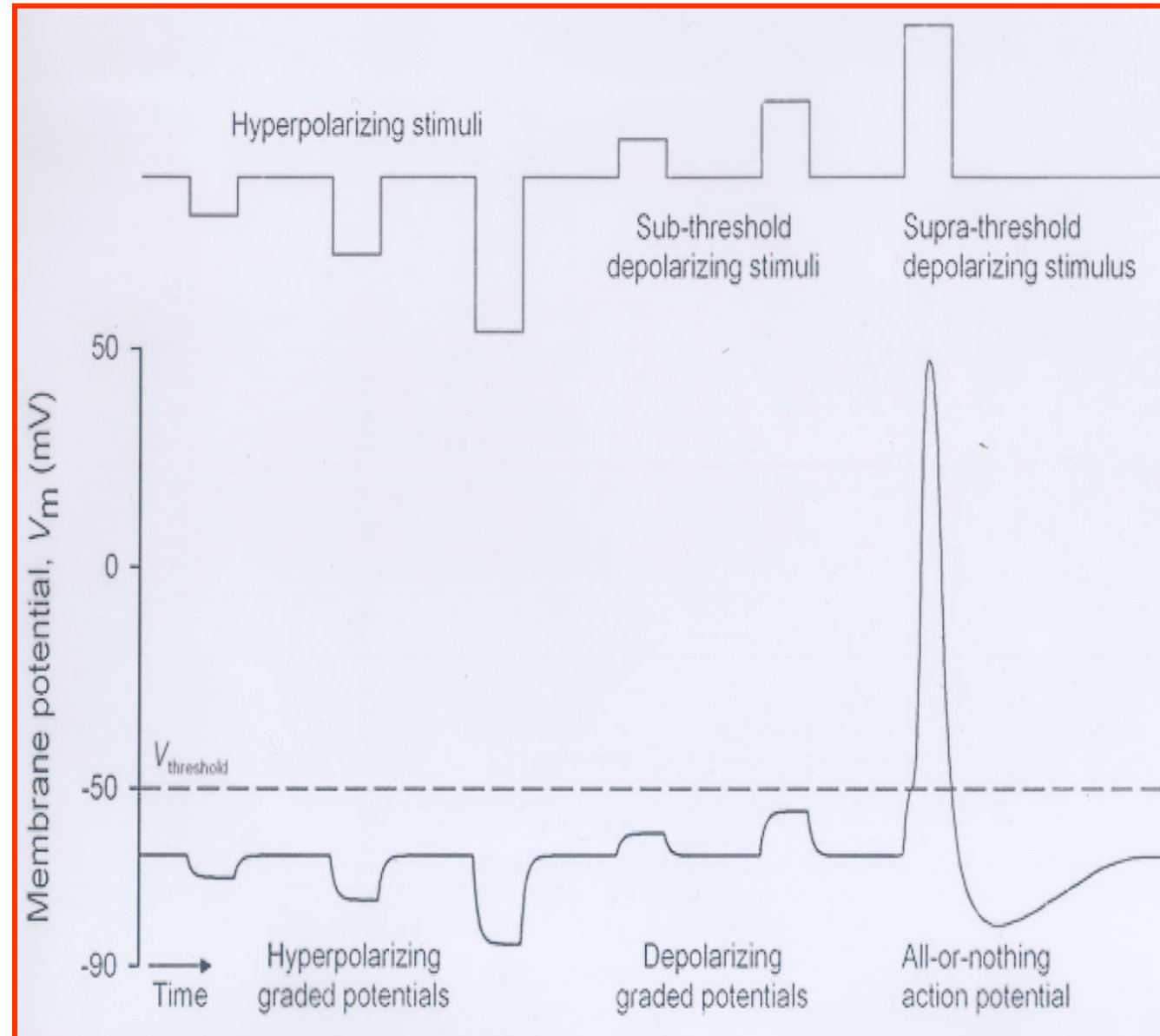
- Varies from neuron to neuron (10-20mV $\rightarrow$ -70mV $\rightarrow$ -55mV)
 - $<$ Threshold $\rightarrow$ passive response
 - $>$ Threshold $\rightarrow$ active response $\rightarrow$ depolarization
- **Significance** $\rightarrow$ small random depolarization of the membrane do not generate AP
- **Amplitude** of the AP is **independent of the stimulus intensity**

v. Strength of stimuli

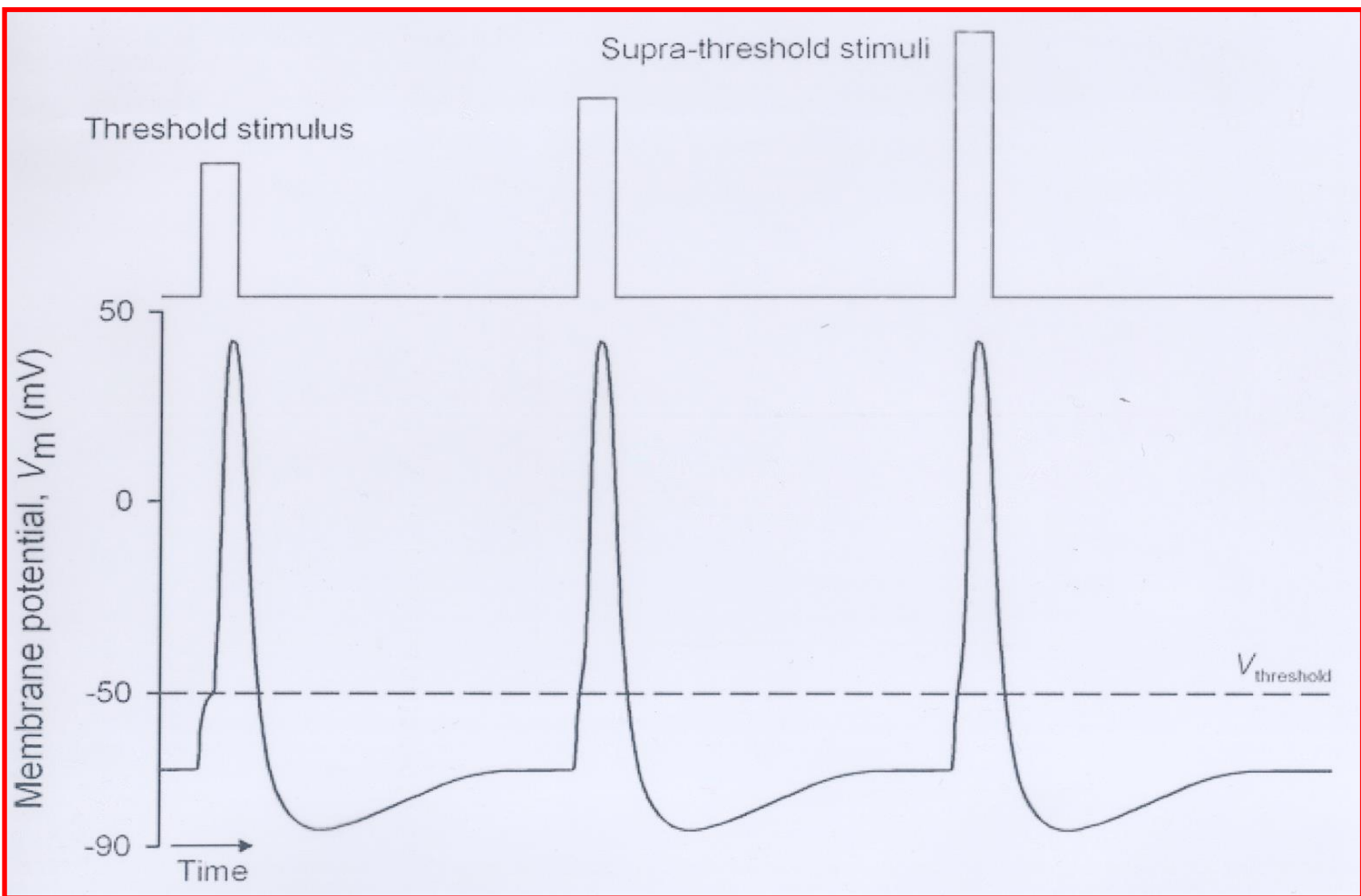
- *Subthreshold*



- *Threshold Stimuli*



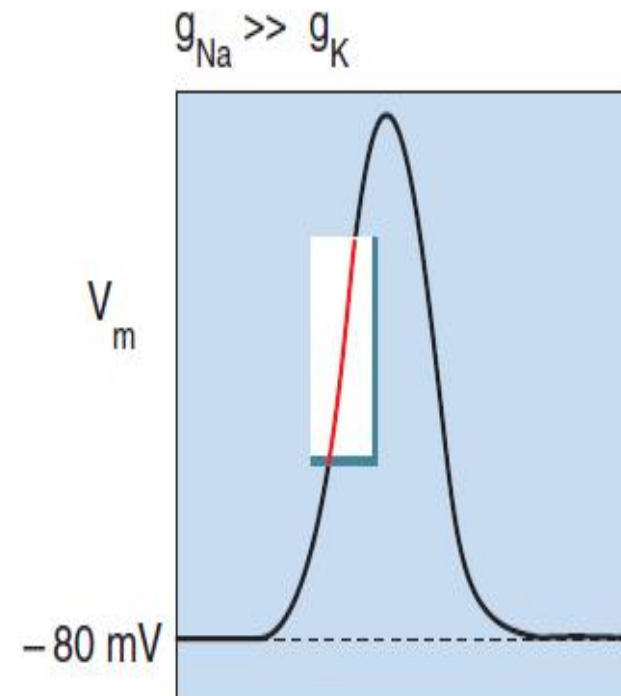
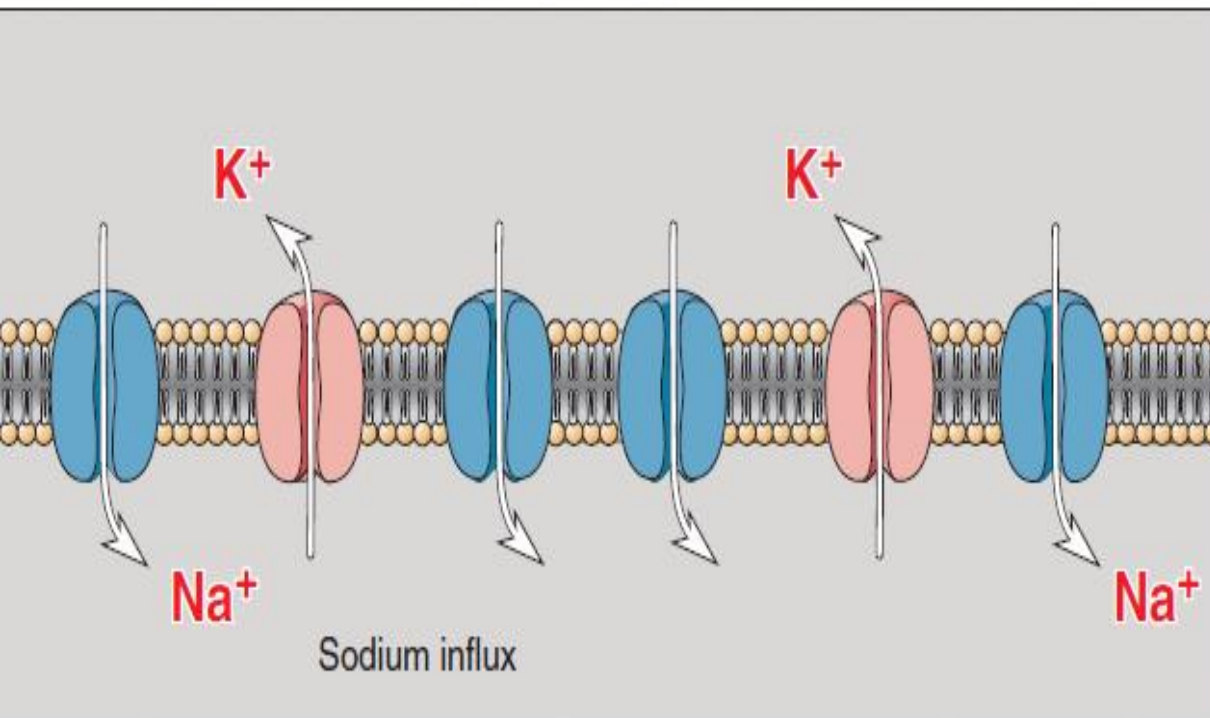
Suprathreshold stimuli



- ***Strength-latency relationship:***
 - **Latency** → the time delay from the onset of stimulus to the peak of AP
 - The stronger the stimulus the shorter the delay between stimulus application and generation of AP

2. Depolarization Phase/ upstroke

- $\uparrow g_{\text{Na}^+} \rightarrow$ flow of Na^+ into the cell
- **m, h** gates opened



- Voltage-gated Na^+ channels can exist in three conformations:-

1. Resting state:

- Closed but capable of opening
- Activation gate closed, inactivation gate open

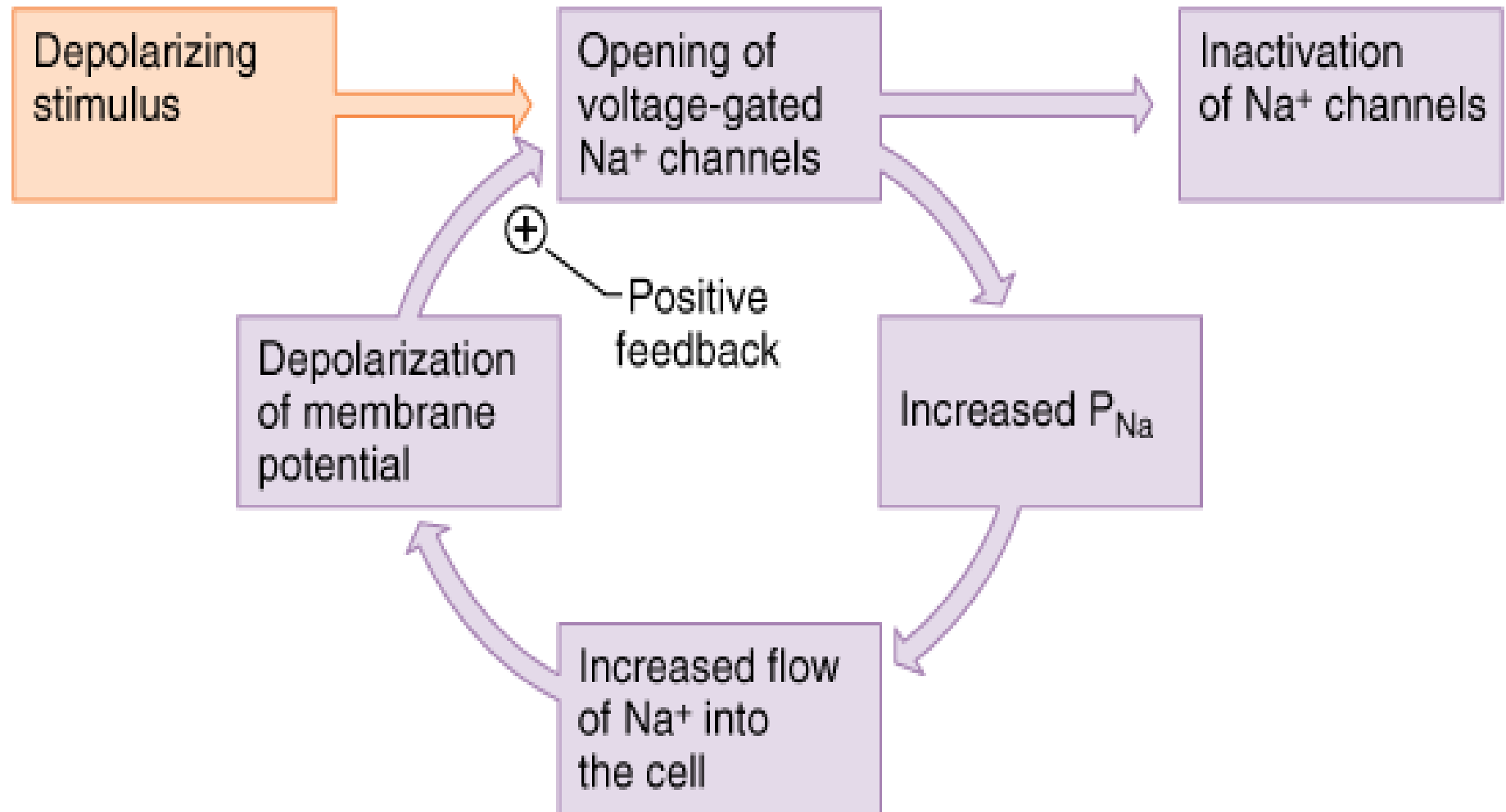
2. Activated state:

- Both gates open

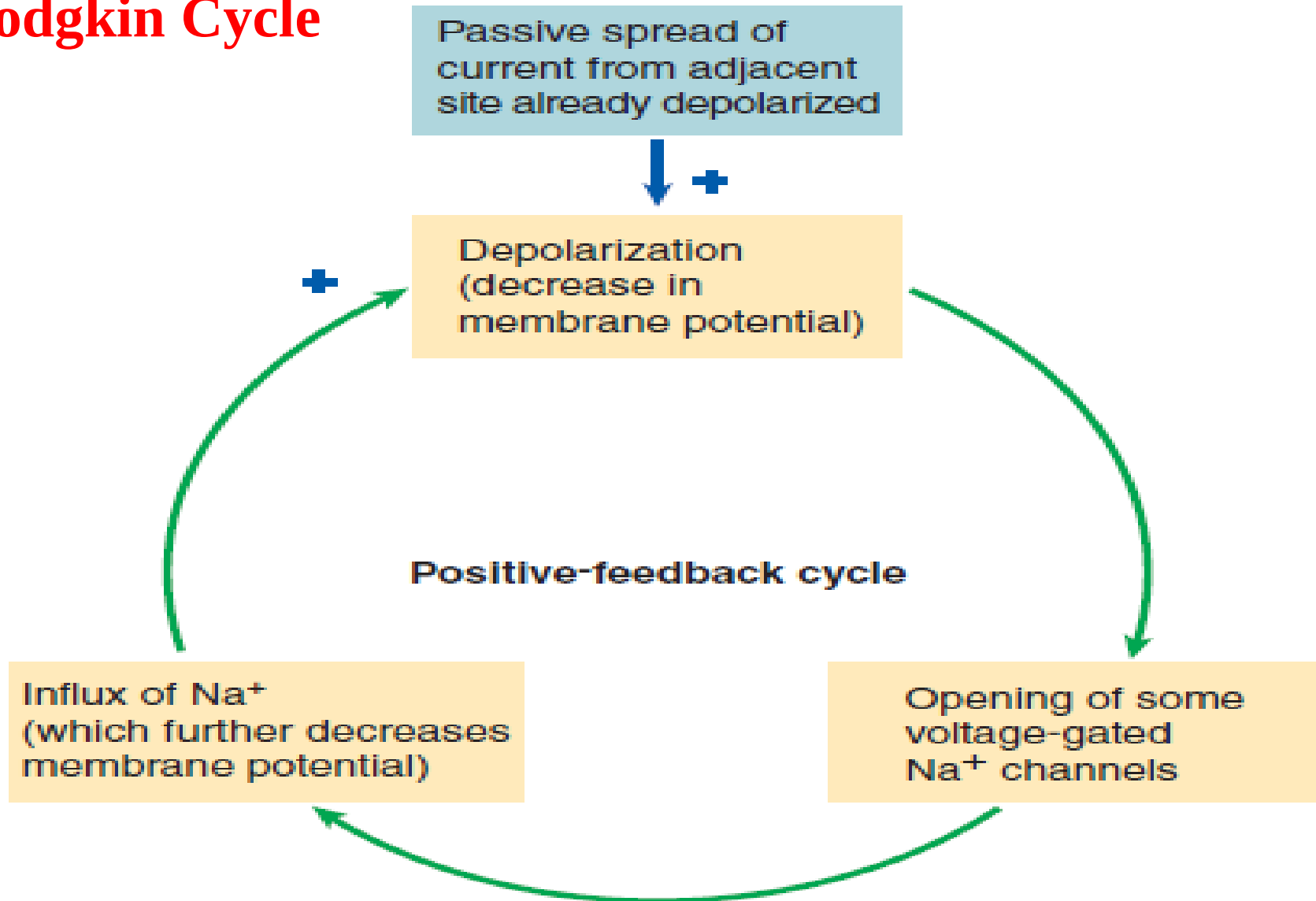
3. Inactivated state:

- Closed and not capable of opening
- Activation gate open, inactivation gate closed

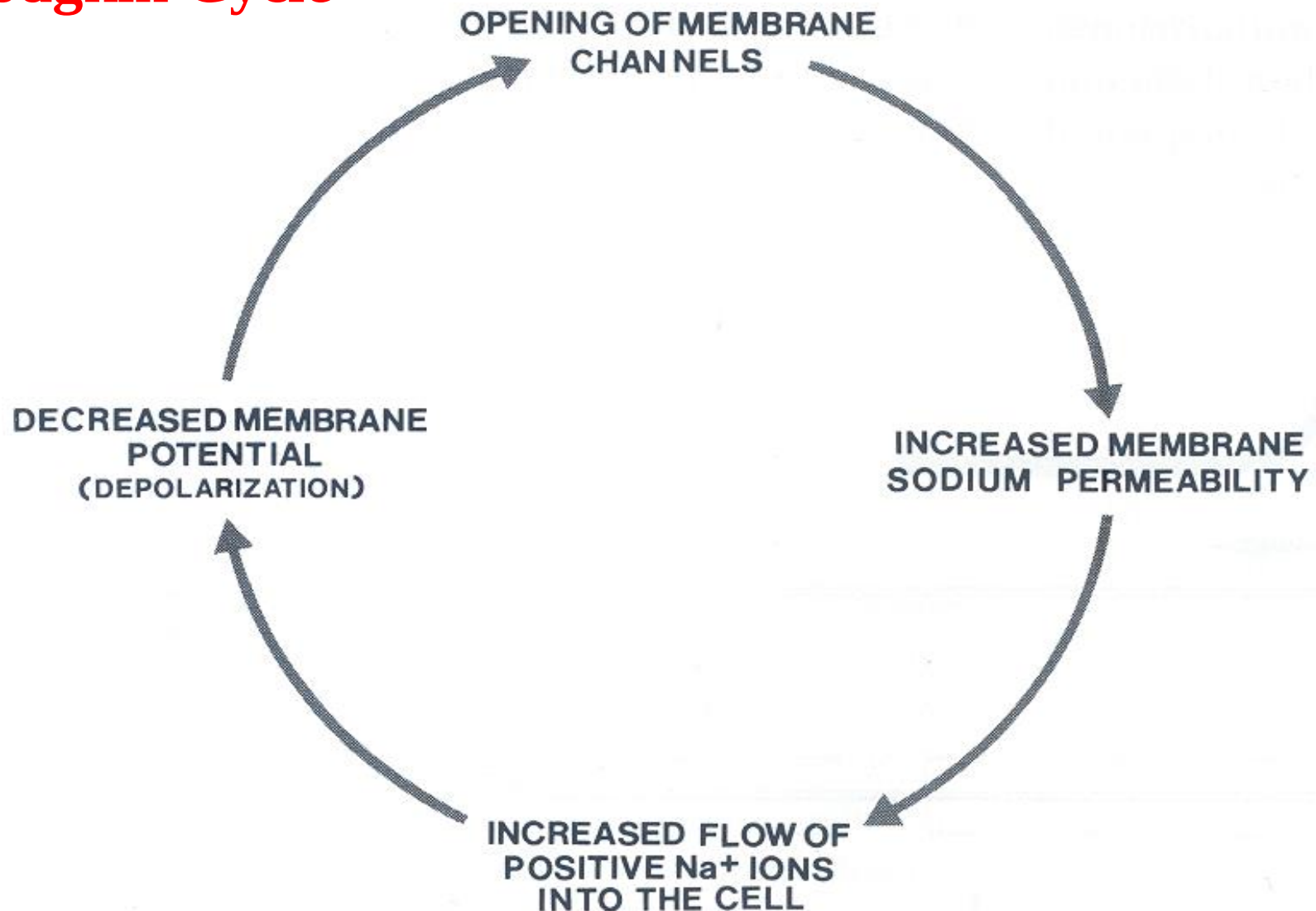
- Sodium channels exert positive feedback on membrane potential



Hodgkin Cycle



Hodgkin Cycle

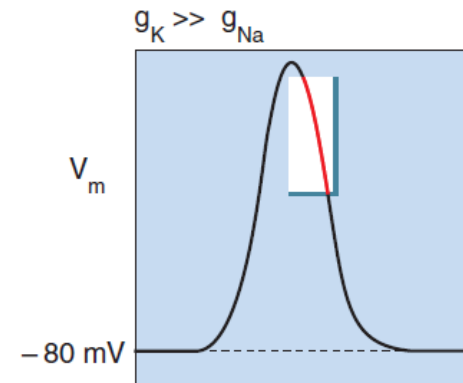
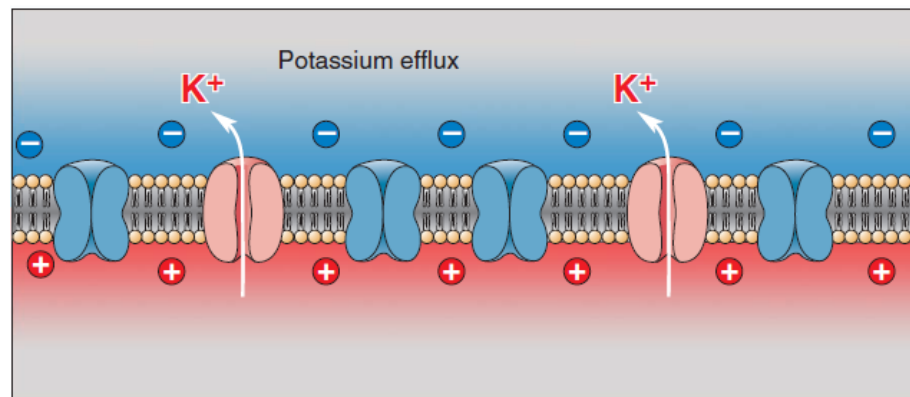


3. Overshoot/peak of the action potential

- Portion of the AP during which the membrane is positive
- Magnitude: 30 to 40mV
- Approaching E_{Na}^{+}

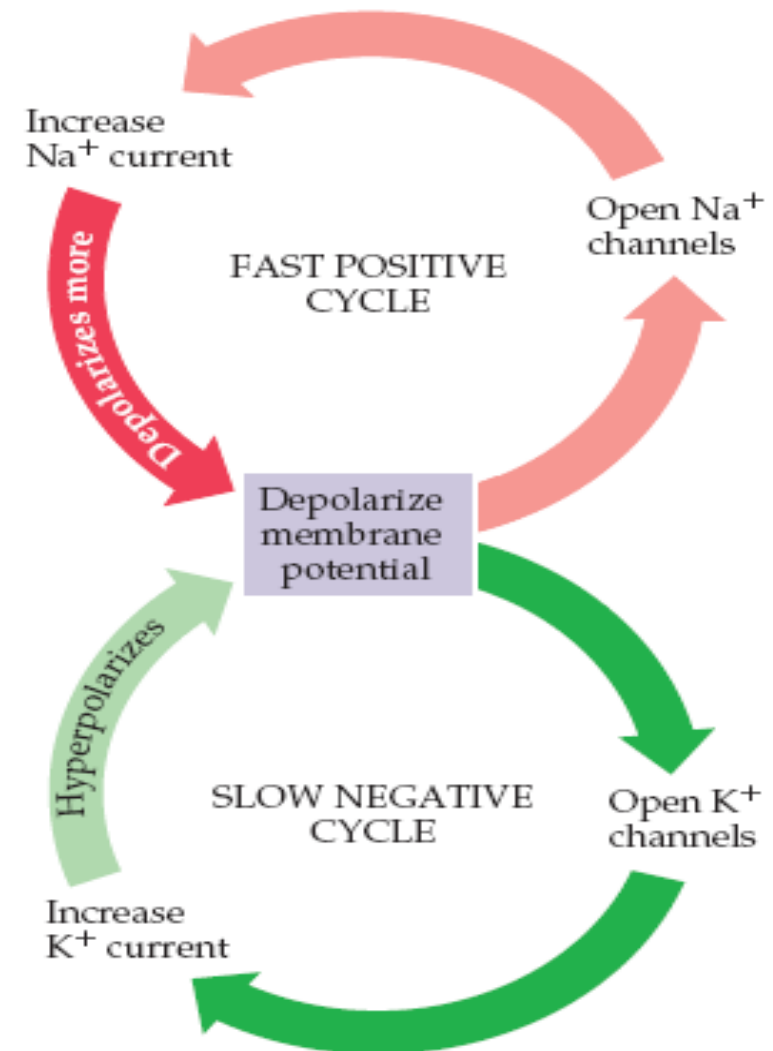
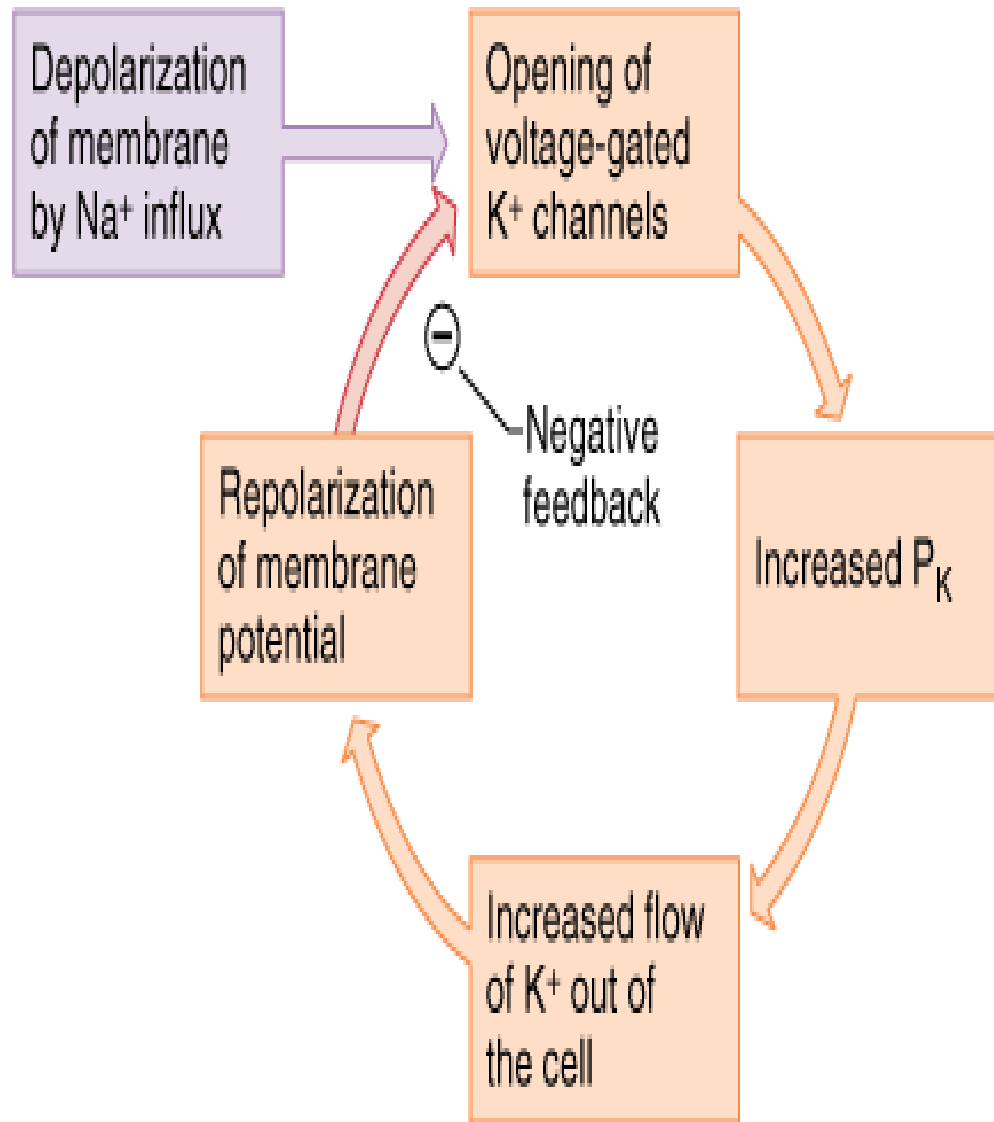
4. Repolarization/Downstroke

i. Rapid return of the membrane towards its RMP.



ii. **Reasons:**

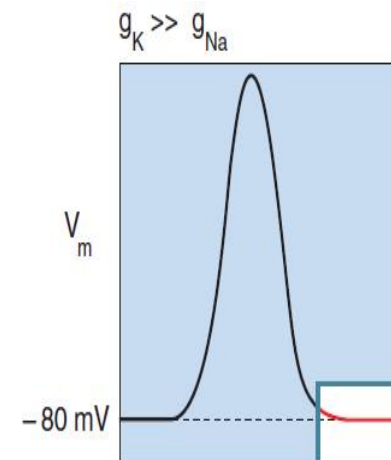
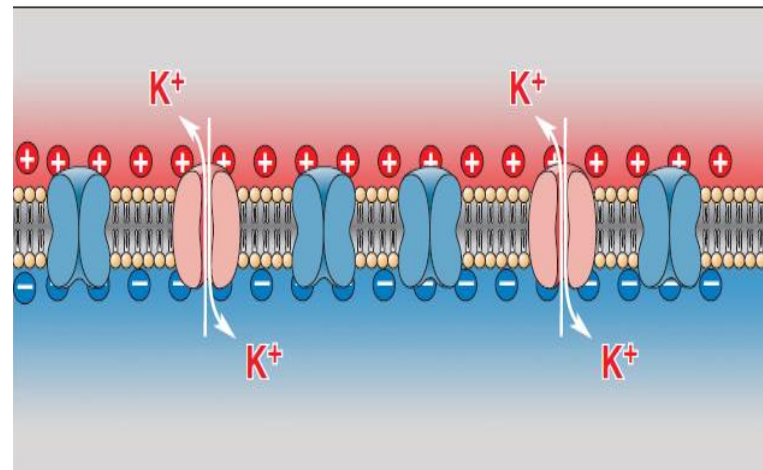
- Time-limited nature of Na^+ permeability (g_{Na}^+ is short-lived)
- Closure of h-gate of $Na^+ \rightarrow Na^+$ inactivation
- $\uparrow gK^+$ (delayed opening of K^+ channels, n-gates $\rightarrow$ open, activated)

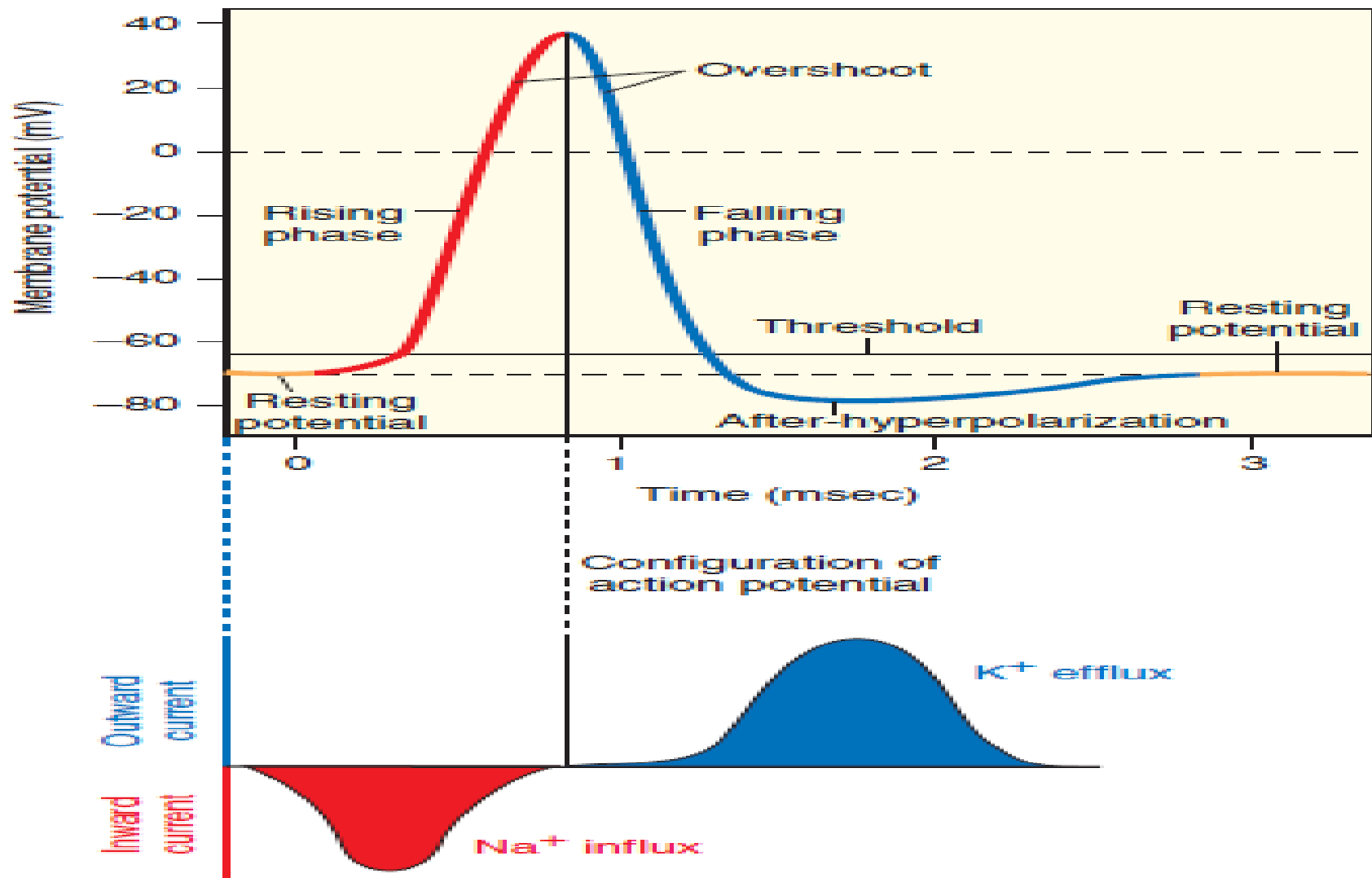


Potassium channels exert negative feedback.

5. Afterpotentials/undershoot

- Membrane becomes more negative than its RMP at the end of the action potential.
- ($-70\text{mV} \rightarrow -72/-73\text{mV}$, 40ms)
- n gates opened $\rightarrow \uparrow \text{K}^+$ out of the cell $\rightarrow \uparrow$ inside negativity
 $\rightarrow \downarrow$ excitability.





Phases of an AP and Inward and outward currents

e. Refractoriness/Refractory period

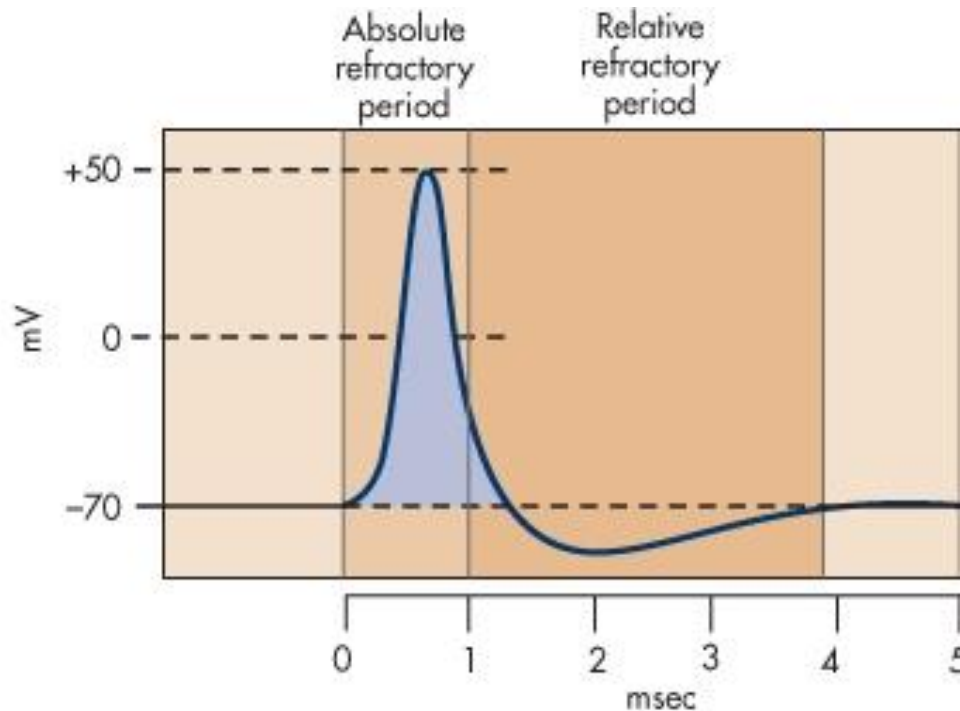
i. Def. ~ an interval during which it is more difficult to elicit an AP.

($\because$ voltage and time-dependent nature of gating particles) .

ii. Types:

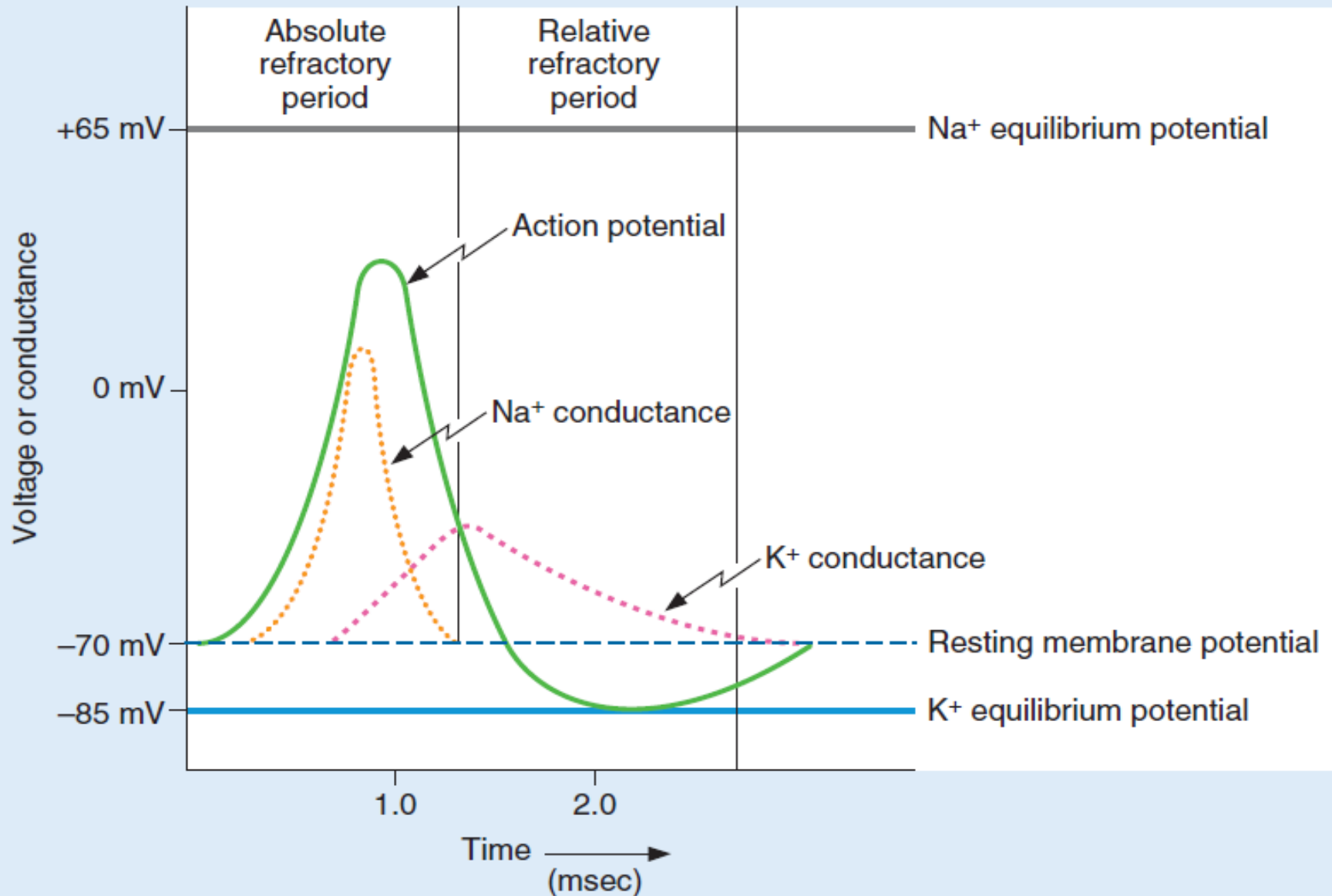
a. *Absolute refractory period*

b. *Relative refractory period*



a. Absolute Refractory Period:

- Another AP can not be elicited, regardless of the strength of the stimulus
- Begins at the start of the upstroke and extends into the down stroke
 - During the upstroke phase → *m* gates are opening
 - During the early portion of the down stroke phase → Na^+ channels are inactivating by *h* gates
- Ends when the number of inactivated Na^+ channels is reduced by repolarization



b. Relative Refractory Period

- A second AP can be elicited if the stimulus is adequate
- Stimulus must be greater than the threshold stimulus (suprathreshold)
 - Na^+ channels are still inactivated
 - More K^+ channels than normal are still open

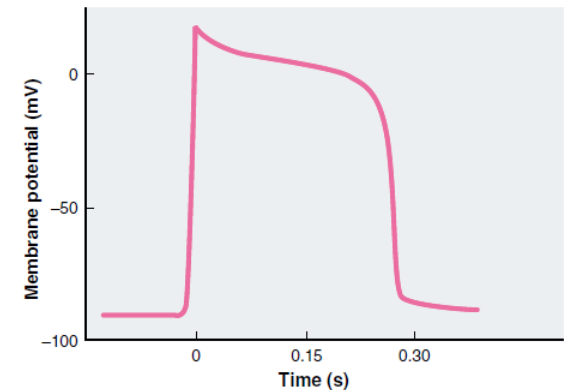
Action Potentials with Plateau

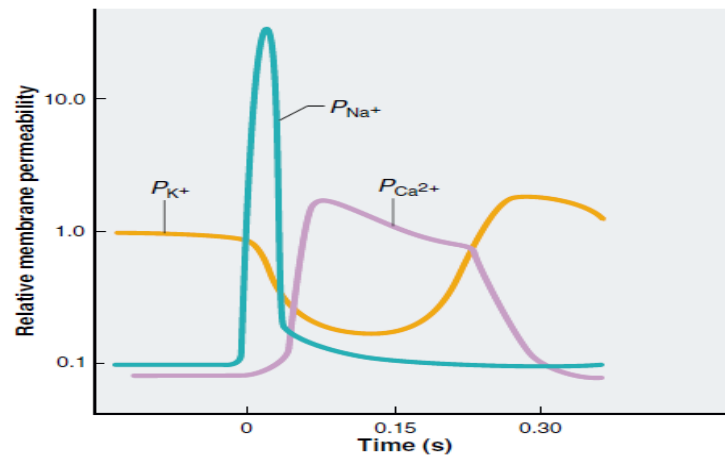
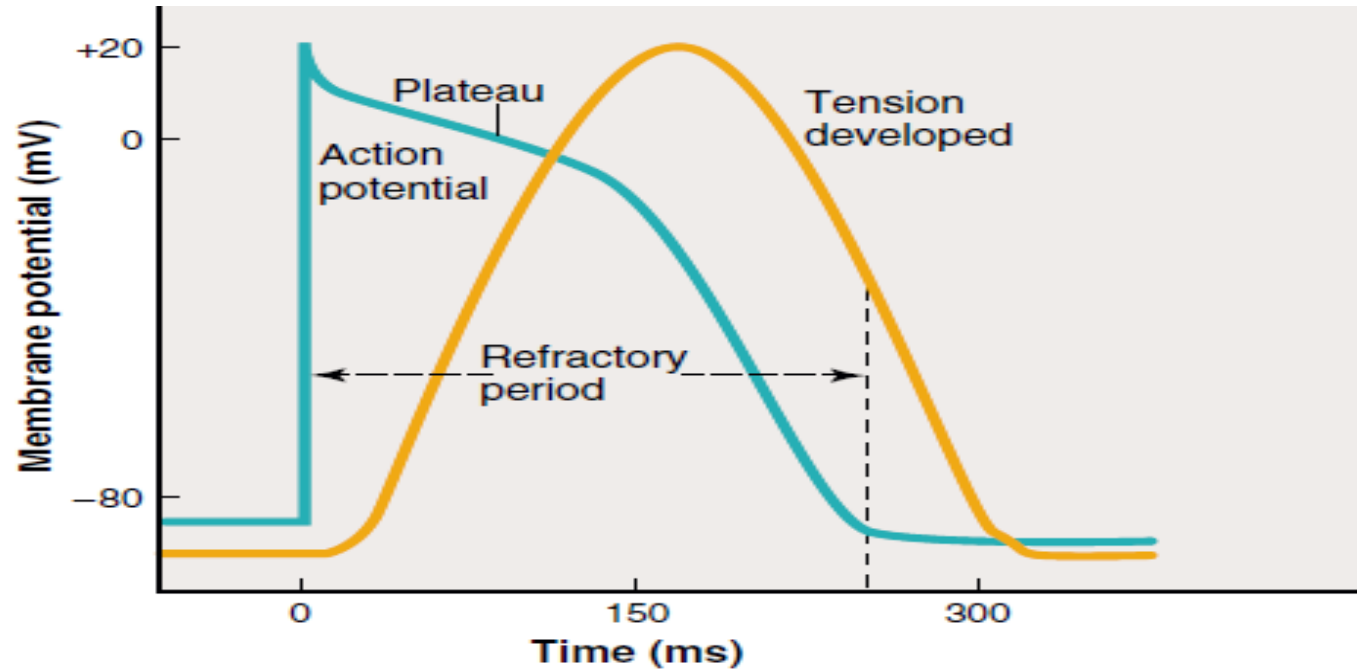
- Plateau prolongs the period of depolarization

Reasons :

- Fast channels: Voltage-gated Na^+ channels $\rightarrow$ spike portion of AP

Slow channels: Voltage-gated Ca^{2+} channels (L-type) $\rightarrow$
 Ca^{2+} - Na^+ channels $\rightarrow \uparrow g_{\text{Ca}^{2+}} \rightarrow$ plateau
- Voltage-gated K^+ channels: slow/delayed opening.



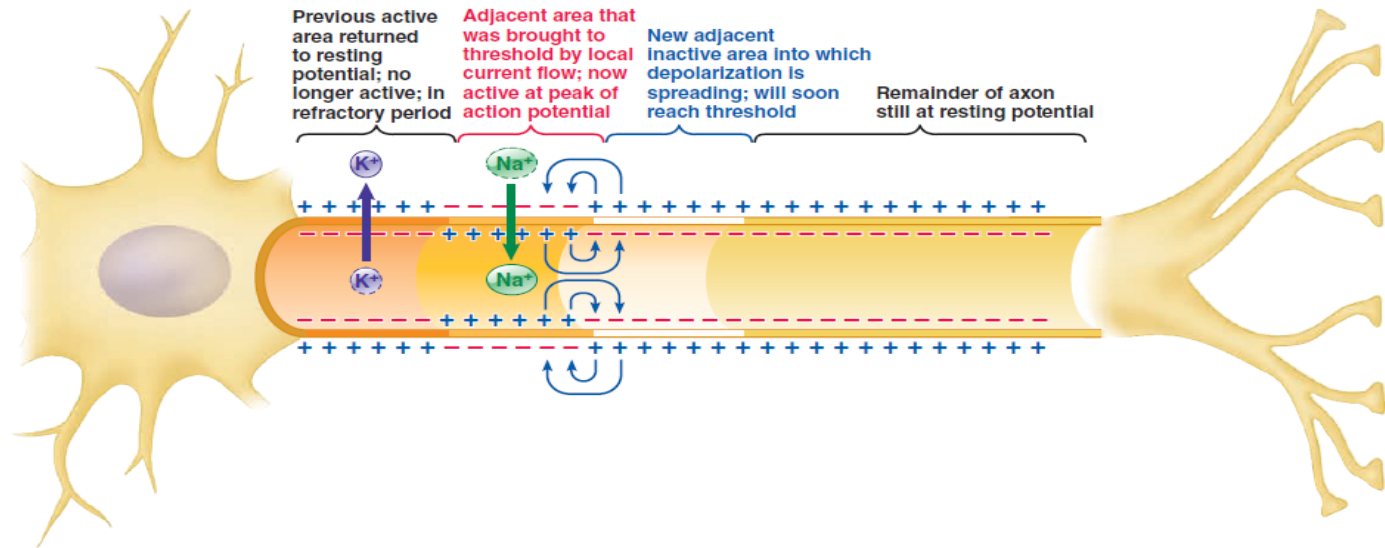
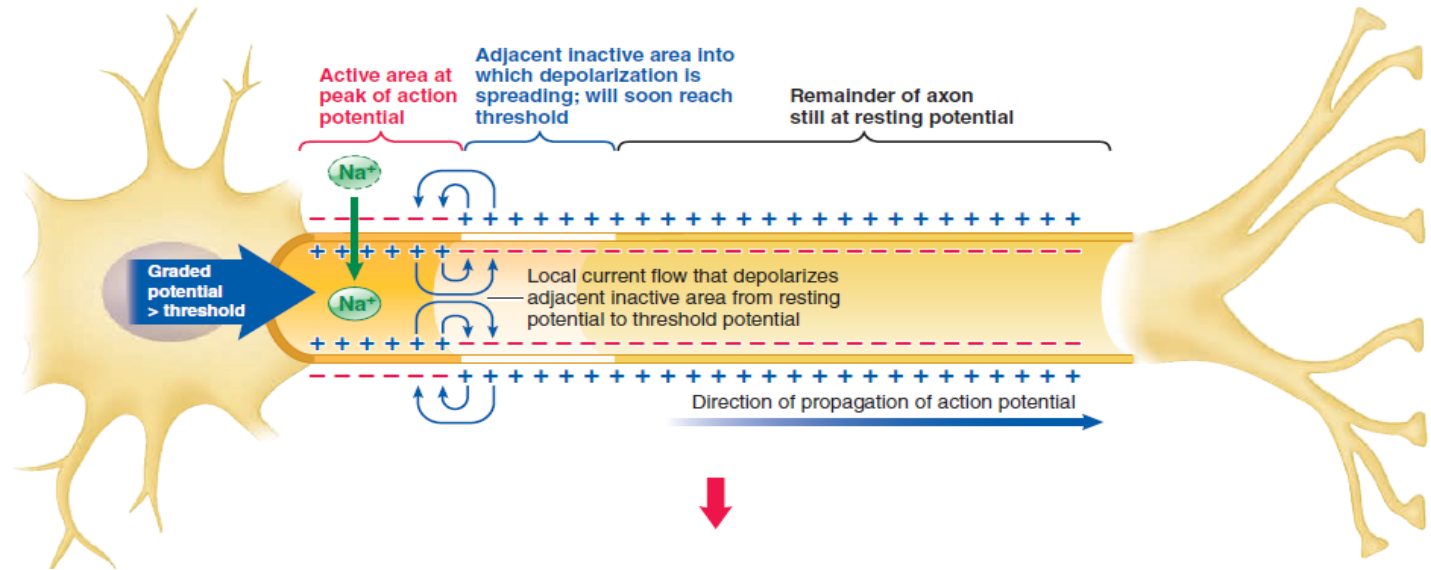


f. Propagation of Action Potential

Types:

i. Cable conduction/Continuous/Sweeping/Contiguous conduction

- Occurs in Unmyelinated nerve
- Speed of AP $\propto$ diameter of the fiber
 - Slow $\sim 1\text{m/s}$



Cable Conduction

ii. Saltatory Conduction

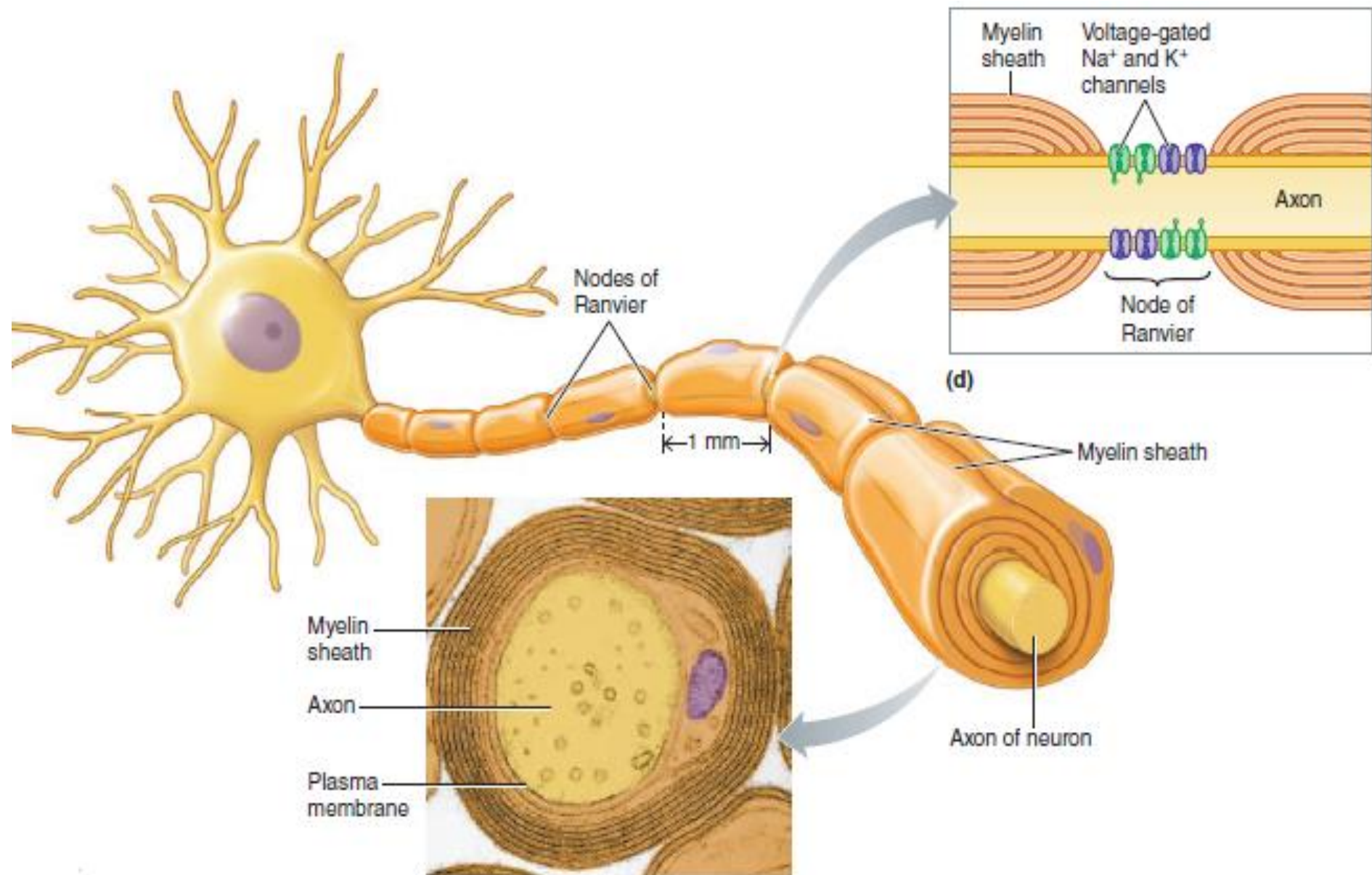
- Occurs in myelinated nerves

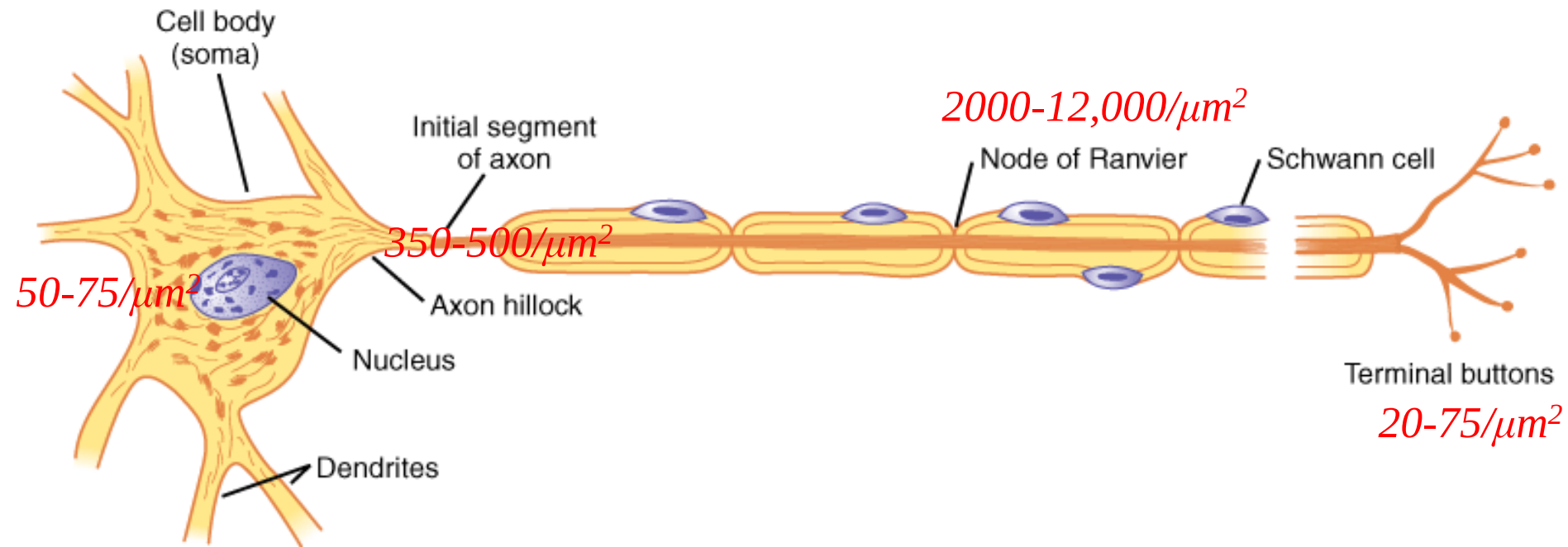
1. Myelin sheath:

Schwann cells → *PNS*

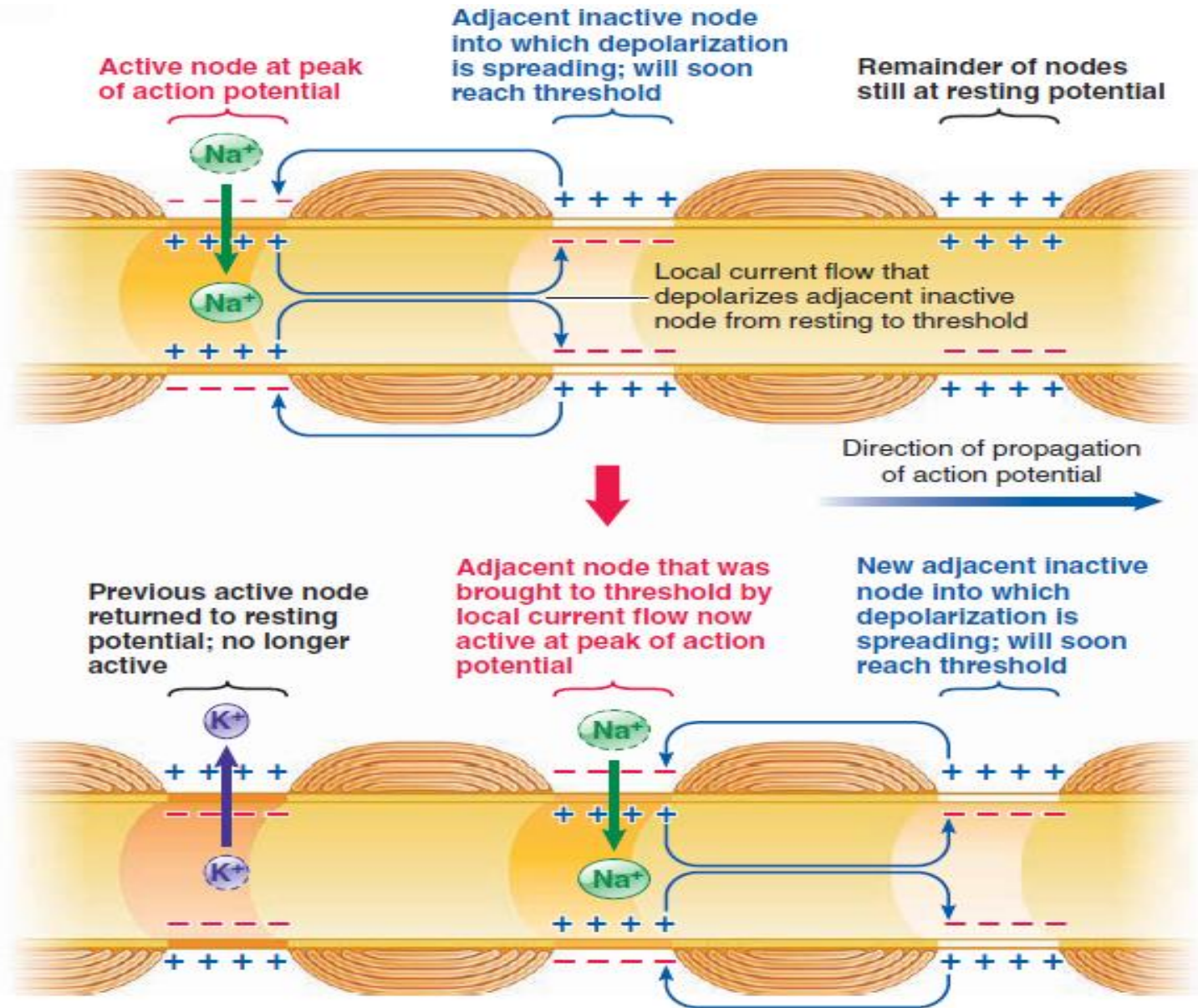
Oligodendroglia → *CNS*

2. Nodes of Ranvier

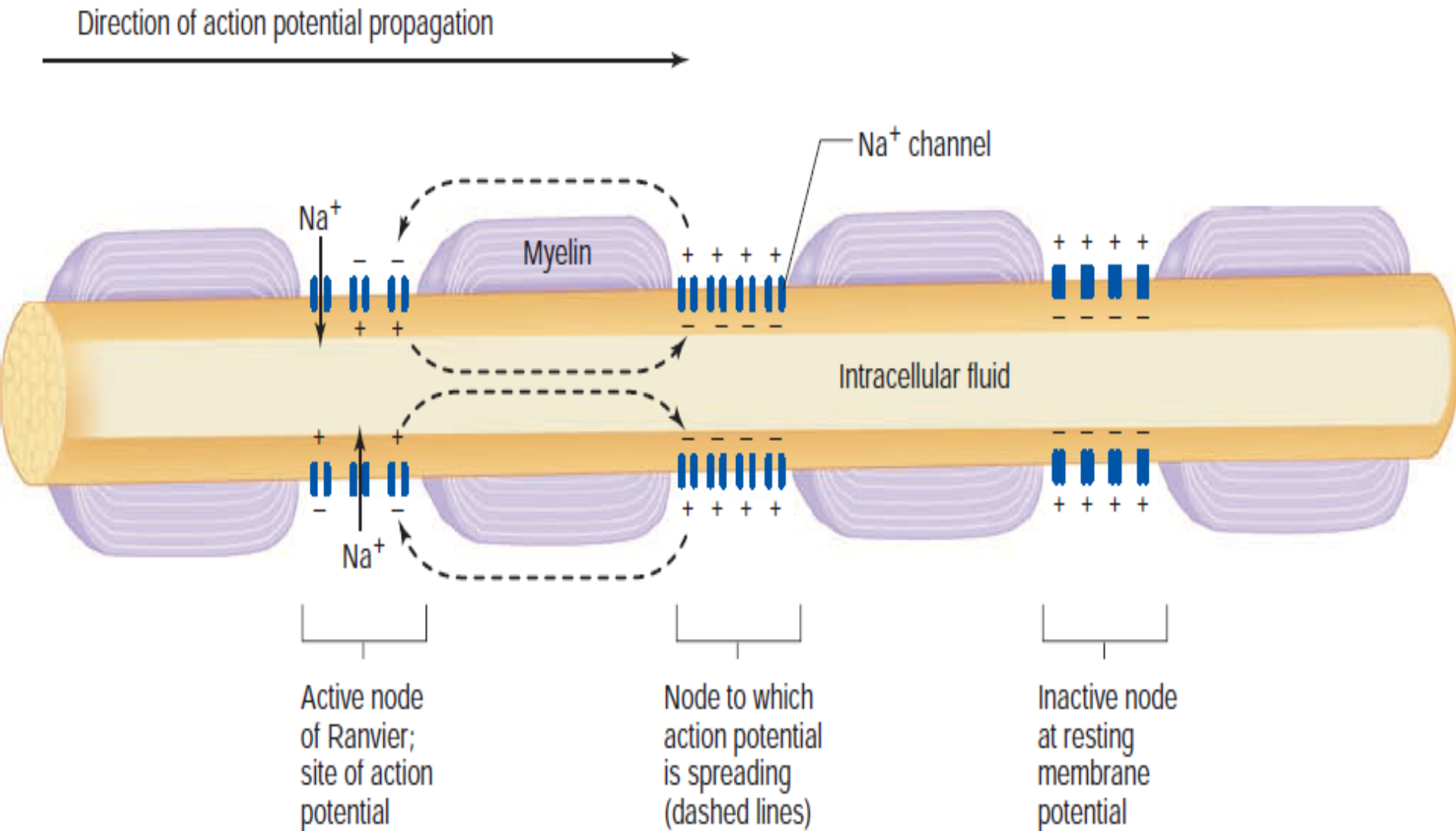




Unmyelinated neuron: $110/\mu\text{m}^2$



Saltatory conduction



Saltatory conduction

iii. *Significance:*

i. *Myelin sheath:* high resistance + lower capacitance.

ii. Fast $\sim$ (6-120m/s)

iii. *Conservation of metabolic energy* (efficient)

- Factors affecting AP propagation

- a. *Type of fiber*

- **Velocity $\propto$ myelination**

- b. *Geometry of the fiber:*

- **Velocity $\propto$ diameter of the fiber**

- c. *Temperature*

- $\downarrow^{\circ}\text{C} \rightarrow \downarrow$ rate of change of permeability

1. Metabolic

a. *pH* : Acidosis $\rightarrow$ $\downarrow$ neuronal excitability (depression)
 Alkalosis $\rightarrow$ $\uparrow$ neuronal excitability

2. Mechanical

a. *Hypoxia*: $\downarrow \text{PO}_2 \rightarrow \downarrow \text{neuronal excitability}$
b. *Pressure*: $\downarrow \text{Excitability} \rightarrow \text{Conduction block}$

3. Drugs and Chemicals

- a. Caffeine (Coffee) → ↑ neuronal excitability by decreasing the threshold potential
- b. Theophylline (Tea) → same mechanism
- c. Strychnine → ↑ neuronal excitability by inhibiting the inhibitory neurotransmitter, Glycine
- d. Anesthetics (Procaine, Lidocaine, Tetracaine, benzocaine) → ↓ excitability
- e. Temperature:
 - Warming → ↑ excitability
 - Cooling → ↓ excitability

vi. Inorganic Ions

- a. Na^+ free solution $\rightarrow$ no nerve impulse
- b. K^+ :
 - $\downarrow \text{K}^+ \rightarrow \downarrow$ excitability
 - $\uparrow \text{K}^+ \rightarrow \uparrow$ excitability
- c. $\uparrow \text{Ca}^{2+} \rightarrow \downarrow$ membrane permeability (*resists Na influx*) $\rightarrow \downarrow$ excitability
 $\downarrow \text{Ca}^{2+} \rightarrow \uparrow$ permeability of Na^+ channels $\rightarrow \uparrow$ excitability